



Theories of Evolution

THEORIES OF EVOLUTION DOMINATED INTELLECTUAL DEBATE IN Europe and America in the latter half of the 19th century, especially after the publication of Charles Darwin's *On the Origin of Species by Means of Natural Selection* (1859). Although religious authorities resisted, natural scientists and the educated public generally embraced theories of evolution: Such theories often represented human progress—or, at least, white, male, and Western human progress—as a triumph of the “survival of the fittest.”

Early Greek thinkers such as Empedocles had advanced theories of the evolution of biological species. However, most scholars during the medieval period had accepted the Aristotelian account of an immutable and hierarchical natural order, or “scala naturae.” Such an account not only sustained the popular conception of a purposive natural order created by a benevolent God, but also conveniently supported the notion of an immutable social hierarchy governed by kings, bishops, and the aristocracy. This account was generally accepted until the late 18th century and—by a good many theorists—beyond. The Enlightenment theories of social development and change advanced by Helvetius and Adam Ferguson (1723–1816) presupposed a fixed human nature, and Gall's phrenology presupposed a more or less fixed hierarchy of neurological function (Young, 1990).

Nineteenth-century evolutionary theorists abandoned the notion of an immutable “great chain of being” (Lovejoy, 1936) and developed explanations of the accepted fact of species change. They generally represented evolution as a process of progressive development toward a hierarchical natural order. Jean-Baptiste Lamarck (1744–1829) and Herbert Spencer (1820–1903) advanced theories that replaced the extrinsic teleology of a divinely created natural order with the intrinsic teleology of progressive development toward a natural order. Charles Darwin (1809–1882) was the exception. His rigorously materialist theory of evolution by natural selection treated evolution as a purely mechanistic process with no extrinsic or intrinsic purpose.

EARLY EVOLUTIONARY THEORIES

Theories of organic evolution began to resurface in the late 18th century (earlier anticipations are to be found in Leibniz and Kant). The English physician Erasmus Darwin (1731–1802), the grandfather of Charles Darwin, advanced a theory of the evolution of animal traits in *Zoonomia* (1794–1796), later popularized in his poem *The Temple of Nature* (1803). His theory was an extension of Hartley’s associationist psychology. Darwin generalized traditional empiricist explanations of the development of individual psychology to the evolutionary development of species by claiming that learned associations and habits engender modifications of the nervous system that are passed on to future generations of a species. Darwin was a committed materialist, who dismissed the notion of an autonomous mental realm as a “ghost story.” According to Darwin’s **fluid materialism**, the electrical nature of the nervous system is the basis of life and mind. Darwin’s work was published in the decades following the excesses of the French Revolution and attracted the same degree of odium as La Mettrie’s *Man Machine*.

Jean-Baptiste Lamarck: The Inheritance of Acquired Characteristics

The evolutionary theory of the French naturalist Jean-Baptiste Lamarck (1744–1829) received a similar reception. Like Darwin and La Mettrie, Lamarck stressed the material continuity of animal species and treated humankind as the most complex form of animal life. He treated evolution as a natural progression from simpler to more complex forms of biological organization, as a consequence of the adaptation of individual organisms to their environments. According to Lamarck, organisms possess an innate drive to perfect themselves and strive to adapt to their environments. According to his principle of the **inheritance of acquired characteristics**, useful modifications that are made to existing organs through increased use or that are developed in response to environmental pressures during the lifetime of an organism tend to be inherited by future offspring (Lamarck, 1809).

To take a familiar example, if some giraffes extend their necks in their effort to reach the leaves on the highest branches of trees, such modifications would tend to be inherited by their offspring, which would account for the characteristically long necks of giraffes. Lamarck believed that the inheritance of such adaptive modifications explained the “transmutation” of species over time, in a linear progression from lower and simpler to higher and more complex organisms.

The principle of the inheritance of acquired characteristics played a significant role in late-19th-century theories of evolution, including Darwin’s theory of evolution by natural selection, but few embraced Lamarck’s own theory. This was largely because of its association with republican and socialist political movements. Political radicals treated his theory as a naturalistic justification for theories of social progress through active “development from below,” particularly through

the emancipation of the working class (Hawkins, 1997). Lamarck died in poverty and disrepute (Boakes, 1984).

A distinctive feature of Lamarck's theory was his teleological assumption of the progressive development of species. A similar assumption about the progressive development of species was made in Robert Chalmers's (1802–1871) anonymously published *Vestiges of the Natural History of Creation* (1844), which linked the development of species to embryonic development. Chalmers, an Edinburgh publisher, maintained that evolution is simply an extension of the growth process, which progresses toward a hierarchical ordering of organic life-forms predetermined by the Creator. At each progressive step, the embryos of a species develop a little further before they mature, increasing the general level of complexity of the species. Despite the fact that Chalmers's theory linked evolutionary development to a divine plan unfolding independently of the adaptation of individual organisms to their environment, *Vestiges* created a sensation when it was published and was subject to a barrage of criticism by both theologians and scientists (Secord, 2001).

Throughout the 19th century many scholars, including many theologians, came to accept the fact of species change, based upon the fossil record. If few rushed to embrace particular theories of evolution, this was because the (rather limited) fossil record was consistent with a variety of different theories of evolution, and the time required for evolutionary change appeared to massively exceed the generally accepted age of Earth, estimated by theologians and naturalists as not more than a few thousand years. Thus, for many, a major impediment to the acceptance of theories of evolution was removed with the publication of *Principles of Geology* (1830–1838) by the English geologist Charles Lyell (1797–1875). Lyell claimed that the forces of geological change are uniform and gradual and operate over extremely long periods of time. He estimated the age of Earth at around 100 million years.

Another work that influenced many evolutionary theorists, notably Herbert Spencer, Charles Darwin, and Alfred Russel Wallace (1823–1913), was *An Essay on the Principle of Population as It Affects the Future Improvement of Society* (1798), by the English political economist Thomas Malthus (1766–1834). He claimed that populations increase geometrically while their food supply increases only arithmetically. When populations outgrow their food supply (as they invariably do), this creates a **struggle for existence**. Spencer, Darwin, and Wallace claimed that in this struggle, only the fittest organisms—that is, those best adapted to their environments—survive and reproduce.

HERBERT SPENCER: EVOLUTION AS A COSMIC PRINCIPLE

Herbert Spencer was the popularizing prophet of evolutionary theory in the 19th century. He developed an account of the evolution of species based upon the inheritance of acquired characteristics and the survival of the fittest, a phrase he

coined that was later adopted by Darwin and Wallace (the co-creator of the theory of evolution by natural selection). It was Spencer who gave the term *evolution* its modern meaning as a description of organic change and established its common association with the notion of progressive change (Bowler, 1975).

Spencer was born in Derby, England. He received only informal and intermittent schooling from his father and uncle, although it seems to have provided him with a critical attitude toward traditional beliefs and a voracious appetite for independent study. He was largely self-taught in biology, physiology, psychology, and philosophy. Spencer worked for some years as an engineer during the railway boom in England in the 1840s, but later moved to London, where he worked as subeditor at the *Economist* and as a freelance journalist. He developed a circle of friends that included Thomas Huxley and George Henry Lewes (1817–1878), the author of the hugely popular *Physiology of Common Life* (1859–1860) and *Biographical History of Philosophy* (1845–1846), who is usually only remembered today for his support of the literary career of his wife Mary Anne Evans (the novelist George Eliot).

Spencer developed an early interest in phrenology, having attended a lecture by Spurzheim as a child. He later became skeptical and turned his attention to philosophy and psychology. His interest in evolution was stimulated by discussions of the “development question” with Lewes (Spencer, 1908). Spencer defended a Lamarckian account of evolution in terms of the inheritance of acquired adaptive characteristics against the embryonic developmental account offered in *Vestiges*. He claimed to have become convinced of the truth of Lamarckian theory as a result of reading Lyell’s critique of it in *Principles of Geology*.

Spencer’s Theory of Evolution

Spencer conceived of evolution as a cosmic force governed by the principle of the conservation of energy, or the “persistence of force.” His theory of evolution was based upon the theory of embryonic development as a process of increased specialization advanced by the German zoologist Karl Ernst von Baer (1792–1876). According to Spencer, the application of persistent force to material bodies leads to their progressive individuation. All forms of evolution involve progressive change from disorganized homogeneity to organized heterogeneity, via the differentiation and integration of the components of physical, chemical, biological, psychological, and social systems.

By this account, disorganized nebular masses evolved into planets and solar systems, material bodies evolved into increasingly more complex organic and living bodies, and primitive hunting and gathering societies evolved into complex industrial societies. The nervous system evolved from the simple forms of primitive organisms to the complex and integrated forms of the mammalian brain.

Spencer explained the evolution of species in terms of the inheritance of acquired adaptations to the environment and the elimination of the poorly adapted through the “survival of the fittest” in the Malthusian struggle for existence:

The average vigor of any race would be diminished did the diseased and feeble habitually survive and propagate; . . . the destruction of such, through failure to fulfill some of the conditions of life, leaves behind those who are able to fulfill the conditions of life, and thus keeps up the average fitness to the conditions of life.

—(1864–1867, 1, p. 531)

Only those organisms that are “well-endowed” through successful adaptation tend to survive and propagate, which ensures the progressive development of species. According to Spencer, this progressive development will continue until humanity attains a perfect congruity of faculties and environmental conditions, guaranteeing individual fulfillment, general happiness, and social harmony:

Finally all excess and all deficiency must disappear; that is, all unfitness must disappear; that is, all imperfection must disappear.

Thus the ultimate development of the ideal man is logically certain. . . . Humanity must in the end become completely adapted to its conditions. . . .

Progress, therefore, is not an accident but a necessity.

—(1851, pp. 64–65)

Spencer, like Lamarck, was committed to a progressive developmental account of evolution based upon the inheritance of acquired characteristics, although the two positions are conceptually independent. The author of *Vestiges* had made a case for the progressive development of species independently of individual environmental adaptation, and Darwin accepted the inheritance of acquired characteristics but denied that evolution is a progressive developmental process.

After the publication of Darwin's *Origin of Species*, Spencer incorporated the mechanism of natural selection within his general theory of evolution (and was somewhat chagrined that he had not thought of the idea himself). However, he never really embraced Darwin's most radical idea that the adaptation of species to environmental conditions occurs through natural selection operating on minor chance variations in the characteristics of organisms. Although he accepted the mechanism of natural selection, he claimed that it operated mainly on lower and simpler vegetative and animal species and that the evolution of higher and more complex animal species is governed by the inheritance of acquired adaptive characteristics:

Natural selection, or survival of the fittest, is almost exclusively operative throughout the vegetal world or throughout the lower animal world, characterized by relative passivity. But with the ascent to higher types of animals, its effects are in increasing

degrees involved with those produced by inheritance of acquired characters; until, in animals of complex structures, inheritance of acquired characters becomes an important, if not the chief, cause of evolution.

—(1893, p. 45)

Spencer's theory, like Lamarck's, presupposed a model of **soft heredity**, according to which the mechanism of inheritance through biological reproduction can be influenced by the life history of organisms: Each new generation of developing offspring somehow "remembers" the experience of its parents and more distant ancestors (Bowler, 1989). Such a model allowed for the adaptive learning experiences of organisms to be "impressed" upon the embryonic development of their offspring.

Consequently, Spencer rejected the theory of the germ-plasm advocated by the biologist August Weismann (1834–1914), since it presupposed a model of **hard heredity**, according to which the mechanism of inheritance through biological reproduction is independent of the life history of organisms. According to Weismann (1893a), the germ-plasm is transmitted to offspring and controls embryonic development, but changes to adult organisms do not affect their germ-plasm, which is isolated in their reproductive cells. By this account, the adaptive characteristics that organisms develop in response to their environment during their lifetime cannot be transmitted to their offspring.

Social Darwinism

Social Darwinism is the application of theories of evolution based upon the survival of the fittest to theories of social change and political practice. The term is misleading because it was Spencer who coined the phrase "survival of the fittest" and because Darwin had little to say about social change and carefully avoided political controversy. The exact definition of the term remains a matter of debate, complicated by the fact that it is most often used as a term of abuse (Bannister, 1979). Many have been accused of being social Darwinists, but few have admitted to it (Bowler, 1989).

Spencer's own **laissez-faire** version of the doctrine is probably the best known and certainly the most influential. He maintained that social progress is best assured by leaving biological, psychological, and social evolution to take its natural course. Individuals should be left to fend for themselves in changing environments and suffer the consequences if they fail. He opposed any form of state intervention to relieve the plight of "inferior" creatures afflicted by poverty, unemployment, disease, or insanity, because it would impede the natural progression of evolutionary change:

There cannot be more good done than that of letting social progress go on unhindered; yet an immensity of mischief may be done in the way of disturbing, and distorting and repressing, by policies carried out in pursuit of erroneous conceptions.

—(1876, pp. 401–402)

Spencer protested vigorously against poor relief, state education and medicine, and even government banking and postal systems. He maintained that the struggle for existence must be left to work out its course without “the artificial preservation of those who are least able to take care of themselves” (1874, p. 343) and that publicly funded welfare schemes merely preserve those organisms that are unfit to survive. For Spencer, this protest amounted to something close to a moral crusade. As he starkly put it, all organisms, from the simple amoeba to the most complex industrial society (which Spencer conceived as a complex organism), get what is due to them from evolution:

If they are sufficiently complete to live, they *do* live, and it is well they should live. If they are not sufficiently complete to live, they die, and it is best that they should die.

—(1851, p. 380)

Spencer, who described his form of social Darwinism as “true” liberalism, mounted a sustained attack on utilitarian justifications of legislation and state intervention designed to ensure the greatest amount of happiness for the greatest number of people. He argued that utilitarian theory could not account for individual and racial differences, which Spencer claimed were a natural consequence of the “adaptation of constitution to conditions” (1851, p. 61). He maintained that general happiness could be achieved only by allowing individuals and societies to develop naturally, enabling superior individuals to attain true happiness through the exercise of their faculties, including their developed moral sense (which he rather idealistically supposed would lead men to recognize their mutual interdependence in society).

Spencer’s commitment to a progressive developmental theory of evolution based upon the inheritance of acquired characteristics did not mandate his distinctive form of social Darwinism. Lamarck’s theory was also a progressive developmental theory based upon the inheritance of acquired characteristics, yet Lamarck had suggested that “lower” or “inferior” organisms could improve their place in the hierarchy of nature through their directed effort (which is why his theory appealed to political radicals opposed to established social hierarchies).

Indeed, this was how many interpreted the struggle for existence that was believed to provide the engine of progress in the 19th century. Competition was seen as a stimulus that encouraged everyone to become fitter (Bowler, 1989). Many of Spencer’s own followers represented his social Darwinism as a secularized form of the **Protestant work ethic**: Everyone has a chance to rise in society as long as they make the effort to adapt to changing circumstances, and only those who make the effort deserve to benefit (Moore, 1985). This moralistic position toward the poor and the unemployed was also shared by interventionist utilitarian theorists such as John Stuart Mill, who was concerned about the effects of misdirected charity

upon the idle and degenerate. He believed it protected them from “the disagreeable consequences of their own acts” (1869, p. 304) and prevented them from learning from their own experience (Hawkins, 1997). Spencer’s condemnation of the “dissolute and idle” urban underclass was an attitude common to many in the Victorian era (Himmelfarb, 1984).

Progressive developmental theories of evolution based upon the struggle for existence and the inheritance of acquired characteristics were also sometimes employed to support interventionist social programs. Advocates of so-called **reform Darwinism** argued that governments should introduce programs designed to create social conditions that would encourage individuals to improve themselves (Stocking, 1962, 1968), notably through improved public health and education (Bowler, 1989).

Spencer’s own commitment to laissez-faire social Darwinism predated his commitment to evolution. He wrote an essay against poor relief when he was 16 (Hawkins, 1997), and in the 1840s served as an editor of the *Economist*, a periodical famous for its advocacy of laissez-faire economics.

Evolutionary Psychology

Spencer’s major contribution to psychology was his integration of associationist psychology and evolutionary theory. For Spencer, the evolution of mind represented yet another example of the progression from undifferentiated homogeneity to organized heterogeneity, reflected in a mammalian nervous system that manifested increasingly complex modes of reaction to external stimuli, from basic reflexes and instincts to memory and reasoning.

Spencer’s associationist psychology was largely based upon his reading of Mill and Bain. He treated the principle of association by contiguity as the basis of intelligence in humans and animals:

Hence the growth of intelligence at large depends upon the law, that when any two psychical states occur in immediate succession, an effect is produced such that if the first subsequently recurs there is a certain tendency for the second to follow it.

—(1855, p. 530)

Spencer was one of the earliest theorists to develop a general account of intelligence, which he held to be a function of the quantity and quality of adaptive associations made by organisms (Guilford, 1967). Since he claimed that these are a function of neurophysiological complexity, he maintained that intelligence is a function of brain size.

For Spencer, the principle of association by contiguity was the fundamental principle underlying adaptation, which he characterized in terms of the adjustment

of internal (mental) relations to external (environmental) relations:

The broadest and most complete definition of life will be—The continuous adjustment of internal relations to external relations.

—(1855, p. 374)

Spencer was committed to strong continuity between human and animal psychology and behavior: He maintained that the differences between them are differences in degree rather than kind. Spencer claimed that humans and animals differ only in terms of the complexity of their associative capacities and that the higher cognitive capacities of humans are complex elaborations of their lower or more basic associative capacities, which are also to be found in animals. For Spencer, there was no fundamental difference between reflexive behavior, instinct, memory, and rationality: They were merely increasingly complex forms of association by contiguity.

Spencer's integration of associationist psychology and evolutionary theory enabled him to accommodate instincts within associationist psychology. Earlier theorists had found it difficult to account for instinctual behavior in terms of individual associative learning, but Spencer accommodated instincts by treating them as originally learned adaptive associations that are realized as modifications of nervous constitution and inherited by future generations. He maintained that "reflex and instinctive sequences" are "determined by the experiences of the race of organisms forming its ancestry" and established by "infinite repetition in countless successive generations" (1855, p. 526).

Spencer maintained that memory and reasoning are just more complex forms of association through which inner relations are adapted to outer relations, including Kantian forms of cognition representing relations of causality, space, and time, which he treated as adaptive associations inherited as anticipatory structures of cognition:

Finally, on rising up the human faculties, regarded as organized results of this intercourse between the organism and the environment, there was reached the conclusion that the so called forms of thought are the outcome of the process of perpetually adjusting inner relations to outer relations; fixed relations in the environment producing fixed relations in the mind.

—(1908, p. 547)

Spencer's account of strong continuity between reflexes, instincts, memory, and reasoning in terms of the principle of association by contiguity exerted a powerful influence on later neurophysiologists and psychologists. His account inspired Hughlings Jackson and Ferrier to develop the reflexive sensory-motor theory of the nervous system and anticipated later attempts by American functional and

behaviorist psychologists to explain all forms of human psychology as more complex forms of animal psychology, governed by basic laws of stimulus-response learning identified via the experimental analysis of the behavior of animals.

In developing his account of how “complex reflexes” are inherited as modifications of the nervous system (1855, p. 540), Spencer described the forms of “anticipatory learning” that Pavlov later characterized as conditioned reflexes. He also followed Bain in extending the principle of association by contiguity to associations between behavior and its consequences, anticipating later theories of instrumental conditioning. According to what became known as the Spencer-Bain principle (Boakes, 1984), behaviors that are followed by success, satisfaction, or pleasure tend to be repeated:

On the recurrence of the circumstances, these muscular movements that were followed by success are likely to be repeated; what was at first an accidental combination of motions will now be a combination having considerable probability.

—(1870, p. 545)

Spencer's Impact

The first edition of Spencer's *Principles of Psychology* (1855) attracted little attention, and the later edition (Spencer, 1870) owed much of its success to the publicity surrounding the publication of Darwin's *Origin of Species*. However, the series of books that Spencer published under the rubric of his “systematic philosophy” eventually became best sellers in Europe and America (they sold hundreds of thousands of copies). *First Principles* (1862) was followed by *Principles of Biology* (1864–1867), *Principles of Sociology* (1876), and *Principles of Ethics* (1892), along with the revised *Principles of Psychology* (1870).

Spencer's theories appealed to businessmen and industrialists, who rationalized their ruthless financial practices in terms of the survival of the fittest. John D. Rockefeller and Andrew Carnegie justified their aggressive capitalism as the “working out of a law of nature and a law of God” (Rockefeller, cited in Hofstadter, 1955, p. 45). When Spencer visited America in 1882, Carnegie met him at the dockside. Spencer's national tour was a spectacular success and culminated with a public banquet in New York attended by notable industrialists and financiers of the day. However, Spencer's theories also appealed to the middle-class citizenry of his age, in particular the legions of clerks, bankers, and associated bureaucrats required in the new capitalist economy, who represented themselves as essential cogs in the great evolutionary wheel of industrial and social progress.

Spencer's theories were taught at British and American universities, and William James used his *Principles of Psychology* as a text for the first psychology courses he taught at Harvard, although he complained of Spencer's “hurdy-gurdy monotony.” While Spencer was held in generally high regard by his contemporaries,

members of the intellectual elite such as Mill, Huxley, and Darwin distanced themselves from him. Thomas Carlyle (1795–1881), who was not at all impressed, called Spencer a “perfect vacuum.” His reputation went into rapid decline at the end of the 19th century.

CHARLES DARWIN: EVOLUTION BY NATURAL SELECTION

Charles Darwin’s theory of evolution by natural selection was one of the most significant intellectual accomplishments of the 19th century, matching in significance Newton’s theory of gravitation in the 17th century. It was significant for the same reason. Darwin’s theory represented a triumph of mechanistic explanation over final causal or teleological explanation, in this instance in the realm of biology. Lamarck and Spencer were committed to intrinsic teleology within evolutionary development. Lamarck attributed to individuals an innate drive to perfect themselves through environmental adaptation, and Spencer represented progress as the necessary outcome of evolution. Darwin, by contrast, avoided any suggestion of purpose, perfection, or progress in his rigorously materialist and mechanistic account of the “descent of species.” Like his grandfather Erasmus, Charles Darwin had no time for “ghost stories.” Moreover, his theory of evolution by natural selection was based upon a wealth of accumulated empirical evidence, unlike the speculative theories of Lamarck and Spencer.

Charles Darwin was born in Shrewsbury, England. His father, Robert Darwin, was a wealthy physician, and his mother, Susannah Wedgwood, was the daughter of Josiah Wedgwood (who founded the Wedgwood pottery firm). Darwin did so poorly at school that his father began to despair that he would ever amount to anything. His university career was similarly less than stellar. He began studying medicine at the University of Edinburgh at the age of 16, but could not stomach watching operations. Two years later he transferred to the University of Cambridge to pursue a degree in theology (on his father’s advice), although the prospect of a career as a country clergyman proved no more inspiring than a career in medicine. He graduated with a poor third-class B.A. degree in 1831 (about the equivalent of a 2.0 GPA by contemporary U.S. standards).

Despite his academic underachievement, Darwin was a born naturalist. From his early childhood days he was an avid collector and cataloguer of rocks, shells, and plants. In his autobiography, he claimed that he derived his main pleasure during his days at Cambridge from collecting beetles (Darwin, 1892/1958), which he often stored in his mouth when his hands were busy (Clark, 1986). At Cambridge Darwin made friends with the clergyman and botanist John Stevens Henslow (1796–1861), whom he accompanied on many field trips. Henslow provided Darwin with the opportunity that transformed his life and scientific biology—to travel on H.M.S. *Beagle* on its five-year scientific circumnavigation of the globe.

The Voyage of the *Beagle*

It is commonly supposed that Darwin was engaged as the *Beagle's* naturalist, but this was not the case. The *Beagle* already had an official naturalist, Robert McKormick, who also served as the ship's surgeon, a not uncommon arrangement in the British navy at that time (Gruber, 1969). Darwin was engaged as an unpaid gentleman companion to Robert Fitzroy, the ship's captain (although Darwin's uncle had to pay £2,000 for the privilege). The previous captain of the *Beagle* had shot himself after three years at sea, and Fitzroy was concerned about his own mental stability. Fitzroy advertised among his aristocratic friends for a gentleman companion, ideally a naturalist, to share his dinner table and conversation. Henslow sponsored Darwin for the position, and on December 27, 1831, the *Beagle* set sail from Plymouth with the 23-year-old Darwin on board. The official naturalist, Robert McKormick, quit the ship in disgust in Rio in 1832, in protest against the privileges afforded the aristocratic gentleman-naturalist on board.

The *Beagle's* scientific exploration lasted from 1831 to 1836. The primary mission of the voyage was to survey the coastlines of New Zealand, Australia, and South America. Darwin was able to study and collect biological specimens offshore at many of the landfalls made by the *Beagle*, including the Galápagos Islands (about 600 miles off South America). He became increasingly intrigued by the multitude of species he encountered and their differences in different environments, such as the varieties of tortoises and finches he discovered on the islands of the Galápagos. Throughout the voyage he collected a mass of biogeographical evidence for species change, although he had no idea of the mechanism responsible for it.

One of the books that Darwin took with him on the *Beagle* was Lyell's *Principles of Geology*, which undermined theological and naturalist estimates of the age of the earth at around 3,000 years. Lyell's geological estimate put the age of the earth at about 100 million years, sufficient to support a theory of evolution that postulated small and gradual changes over huge expanses of time, as Darwin's theory later did.

The Theory of Evolution by Natural Selection

On his return to England, Darwin began to develop a theory to account for the evolution of species, or the "descent with modification" of species. According to Darwin (1892/1958), there were two sources of his theory of evolution by natural selection. The first was the established practice of artificial selection by agricultural breeders, who developed desirable characteristics in their animals (such as high quality of fleece in sheep and large body mass in cattle) through selective breeding. Darwin noted how occasionally a harsh winter or drought obliged to produce the outcome desired by breeders, by weeding out the weaker stock. The second was Malthus's *Essay on the Principle of Population*. Darwin recognized that the competition for limited food

resources that Malthus described would *naturally* ensure that variations in inheritable characteristics that were conducive to the survival of a species would tend to be passed on to future generations, whereas those that were not would not.

Darwin came to believe that the **natural selection** of variations in inheritable characteristics could explain the transformation of species over generations:

In October 1838, that is, fifteen months after I had begun my systematic enquiry, I happened to read for amusement Malthus on *Population*, and being well-prepared to appreciate the struggle for existence which everywhere goes on from long continued observation of the habits of animals and plants, it at once struck me that under these circumstances favorable variations would tend to be preserved and unfavorable ones tend to be destroyed. The result of this would be the formation of new species.

—(1892/1958, pp. 42–43)

This was the insight that was published in 1859 as the theory of natural selection in *On the Origin of Species by Means of Natural Selection, or the Preservation of Favored Races in the Struggle for Life*:

As many more individuals of each species are born than can possibly survive; and as, consequently, there is a frequently recurring struggle for existence, it follows that any being, if it vary however slightly in any manner profitable to itself, under the complex and sometimes varying conditions of life, will have a better chance of surviving, and thus be *naturally selected*. From the strong principles of inheritance, any selected variety will tend to propagate its new and modified form.

—(1859, p. 5)

According to Darwin's theory, organisms exhibit chance variations in their characteristics. They produce more offspring than can possibly survive in given environments, creating a struggle for existence. Those chance variations in characteristics that are conducive to the survival of an organism in a given environment are naturally selected, since organisms possessed of these characteristics tend to survive and reproduce, whereas organisms lacking such characteristics tend to die off and fail to reproduce.

In contrast to Lamarck's theory in terms of the inheritance of acquired characteristics, Darwin's theory of evolution by natural selection suggested that those giraffes that happened to have longer necks that enabled them to reach the leaves on the higher trees tended to survive and reproduce, whereas those with shorter necks tended to perish. The natural selection of minor variations in characteristics accounted for the gradual transformation of species over long periods of time.

Darwin's theory of species change through natural selection required that chance variations in characteristics conducive to the survival of an organism would be inherited by future generations of the organism. His own reproductive theory

was based upon the “blending” of adult characteristics, according to which offspring inherit half of the “particles” that pass through the adult parents to their reproductive organs (Darwin, 1868/1896). Darwin’s **blending theory** allowed for a hard theory of heredity, according to which inherited variations are independent of the adaptive adjustments made by individual organisms to their environment during their lifetimes, but also allowed for a soft theory of heredity based upon the inheritance of acquired characteristics (via the modification of reproductive particles). Darwin accepted the inheritance of acquired characteristics, although he tended to downplay its role in evolution (except in the case of human evolution), and always insisted that acquired characteristics are themselves subject to natural selection.

The most significant feature of Darwin’s theory of evolution by natural selection was his suggestion that natural selection operating on chance variations in characteristics is in principle *sufficient* to explain the adaptation of species to their environments and the transmutation of species over time. In this account, there is no need to postulate any innate drive to perfection or inevitable progression. As Aristotle had recognized in his discussion of Empedocles’ theory of evolution, such a process of natural selection operating on chance variations in characteristics could produce adapted species that would appear, but *only appear*, to be purposively designed.

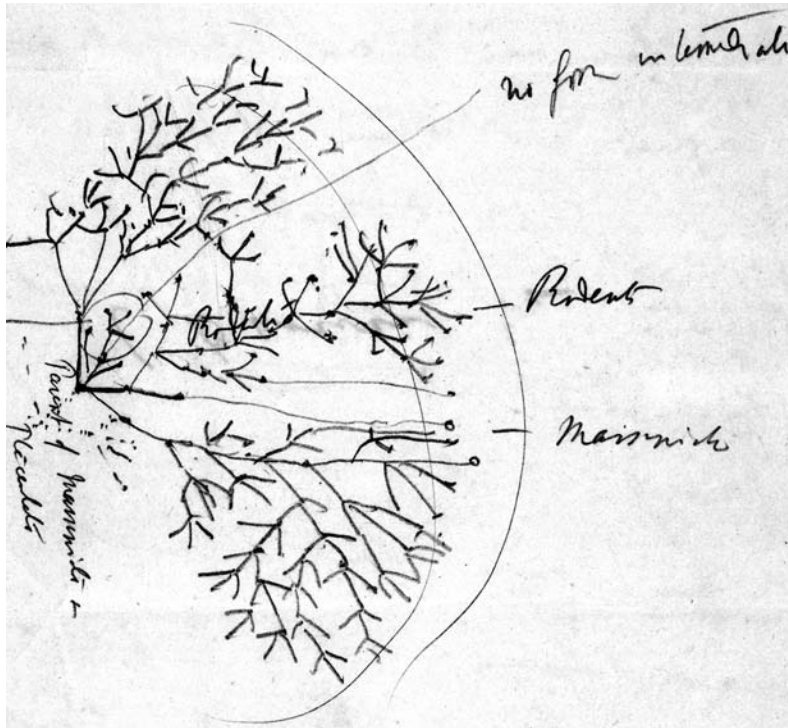
There is no vestige of purpose, perfection, or progress in Darwin’s theory. According to Darwin, evolution is an ongoing mechanistic process, with natural selection operating on chance variations in organisms in changing environments to produce constantly changing species. He claimed that the process of evolution was best represented as an “irregularly branching” tree, and cautioned against the use of terms such as *higher* and *lower* in comparing different species adapted to different environments (Boakes, 1984).

Although he eliminated intrinsic and extrinsic teleology from biological evolution, Darwin did not reject the intrinsic teleology of purposive animal and human behavior. In the case of humans in particular, he insisted that their developed intelligence enables them to consciously and purposively adapt themselves to their environments:

[Man] has great power of adapting his habits to new conditions of life. He invents weapons, tools, and various stratagems, by which he procures food and defends himself. When he migrates into a colder climate he uses clothes, builds sheds, and makes fires; and, by the aid of fire cooks foods otherwise indigestible. He aids his fellow-men in many ways, and anticipates future events. Even at a remote period he practiced some division of labor.

—(1871, p. 158)

The main conclusion of the argument of Darwin’s *Origin of Species* was his denial of the independent creation of a fixed natural hierarchy of species and his



Darwin notebook: evolution as an irregularly branching tree.

claim that present species are descendants of a smaller number of earlier species modified through natural selection:

I am fully convinced that species are not immutable; but that those belonging to what are called the same genera are lineal descendants of some other and generally extinct species, in the same manner as the acknowledged varieties of any one species are the descendants of that species. Furthermore, I am convinced that natural selection has been the most important, but not the exclusive means of modification.

—(1859, p. 6)

Darwin's Delay

Darwin started work on his theory of evolution by natural selection almost as soon as he returned to England in 1836. He began a notebook on the “transmutation” of species in 1837, and probably came to develop his theory around 1840. He sketched versions of it in 1842 and 1844 to be published in the event of his

premature death. Yet he did not publish his theory until 20 years after the voyage of the *Beagle*.

His delay was caused in part by the time spent on his return to England organizing and cataloguing his collection of specimens, accumulated during the voyage of the *Beagle* and forwarded to him by his numerous worldwide correspondents. Darwin published his *Journal of Researches Into the Natural History and Geology of the Countries Visited During the Voyage of the H.M.S. Beagle* in 1839 and spent much of the next decade writing a book on the taxonomy and natural history of barnacles. Another reason for his delay was the more or less constant ill health that he suffered, which restricted him to only a few hours of work each day.

Yet a major reason for his delay may have been his genuine reluctance to publish, given his justified fears of the reaction he anticipated would follow the publication of his theory. Such fears may provide a psychosomatic explanation of Darwin's later ill health (Colp, 1977), although it is also possible that he may have suffered from a tropical disease contracted on his travels (Adler, 1959). During the decades following the voyage of the *Beagle*, Darwin established and consolidated his scientific reputation. He was elected a fellow of the Royal Society in 1839 and a corresponding member of many international scientific societies. Yet he was in no rush to present a rigorously materialistic and mechanistic theory that was bound to provoke a hostile reaction among theologians, the general public, and many naturalists. Indeed, Darwin might never have published his theory in his own lifetime had he not received a copy of a paper from a fellow English naturalist, Alfred Russel Wallace, titled "On the Tendency of Varieties to Depart Indefinitely from the Original Type."

Wallace had spent years gathering biological specimens in the Amazon jungle, but had lost his collection at sea when his ship caught fire and sank on the return voyage to England in 1852 (he was rescued after 10 days in a leaky lifeboat). He renewed his research on the "question of questions" in the Malay Archipelago shortly afterward. During a bout of tropical fever, Wallace suddenly realized that natural selection could serve as the mechanism for evolutionary change and wrote his paper within a few days (Magner, 1994). Darwin immediately recognized that Wallace's theory (which also acknowledged the influence of Malthus) was almost identical to his own. Darwin's first inclination was to cede intellectual priority to Wallace. However, he was persuaded by his friends to have Wallace's paper and a hastily prepared statement of his own theory read at the July 1858 meeting of the Linnean Society. Neither Darwin nor Wallace was present, and the papers were read into the minutes of the society by its president.

The Reception of Darwin's Theory

These initial statements of what became known as the theory of evolution by natural selection aroused little immediate interest, and they were not noted as

particularly significant in the Linnean Society's annual report. The publication of Darwin's *On the Origin of Species by Means of Natural Selection* the following year was a different matter. Darwin's work and reputation were well known, and the publication of his theory was eagerly anticipated (the first print run of 1,250 copies is reputed to have sold on the first day of publication). It immediately became the object of religious and scientific controversy. Wallace graciously ceded priority to Darwin and characterized the theory of evolution by natural selection as "Darwinism" (Magner, 1994).

A famous instance of this controversy was the Oxford meeting of the British Association in 1860, at which Thomas Huxley, who came to be known as "Darwin's bulldog," vigorously championed Darwin's theory against the objections to the "monkey-theory" advanced by Bishop Samuel ("Soapy") Wilberforce. In the course of his critique of Darwin's theory, Wilberforce insultingly inquired of Huxley whether he was descended from an ape on his grandmother or grandfather's side. Huxley responded that he would "rather have a miserable ape for a grandfather" than a man capable of abusing his intellect and resorting to ridicule in serious scientific debate (Bibby, 1959, p. 69). Darwin wrote to Huxley in congratulation:

How durst you attack a live bishop in that fashion? I am quite ashamed of you! Have you no respect for fine lawn sleeves? By jove, you seem to have done it well.

—(cited in Bibby, 1959, p. 70)

However, the Oxford meeting of the British Association was not as dramatic as it was later portrayed (Richards, 1987). Contrary to legend, the now Admiral Fitzroy did not wander around muttering "the book, the book," while holding a Bible above his head, but merely took his turn at the lectern. Yet, for Fitzroy, a staunch believer in the literal biblical account of creation, Darwin turned out to have been an unfortunate choice of gentleman companion to alleviate his suicidal tendencies. Fitzroy blamed himself for indirectly contributing to the development of Darwin's blasphemous theory, and five years later in 1865 he cut his own throat. In that same year Gregor Mendel (1822–1884) identified dominant and recessive factors in heredity, the foundation of the modern theory of genetics and the later neo-Darwinian theory of evolution.

In the United States, Darwin's theory was vigorously attacked by the prominent Harvard biologist Louis Agassiz (1817–1873) in the late 19th century and remained controversial in the 20th century. William Jennings Bryan (1860–1925) and Clarence Darrow (1857–1938) clashed during the Scopes "monkey-trial" in Tennessee in 1925, in which John Thomas Scopes, a high school teacher, was prosecuted and found guilty of the crime of teaching the theory of evolution by natural selection in high school. The Supreme Court struck down state laws banning the teaching of evolution in public schools in 1968, but the controversy continues (Magner, 1994). Even at the beginning of the 21st century, only about

50 percent of Americans believe that humans evolved over millions of years from earlier life forms (according to a 2007 Gallup poll).

Darwin was careful to avoid religious controversy and described himself as an agnostic. Although others were quick to identify *Origin of Species* as a materialist and atheist tract (both theological critics and communist sympathizers such as Karl Marx), Darwin played down these implications (and remained friends with Bishop Wilberforce). Although he denied the literal biblical account of creation and the fixity of species, these positions had already been abandoned or questioned by many scientists and a good number of theologians who accepted the evidence for species change.

Darwin did not deny the existence of a benevolent and intelligent Creator, only a divine role in the descent of species, and talked of the “several powers” that were “originally breathed into a few forms or into one” (1859, p. 490). But he was not hesitant to respond when others suggested a divine role in evolution. The American botanist Asa Gray (1810–1888), who was one of Darwin’s foremost American supporters, suggested that God might have controlled variations in the characteristics of plants and animals to ensure certain desirable evolutionary outcomes. Darwin (1868/1896) responded that such a hypothesis was inconsistent with the evidence from plant and animal variation, a point that Gray eventually conceded (Reed, 1997).

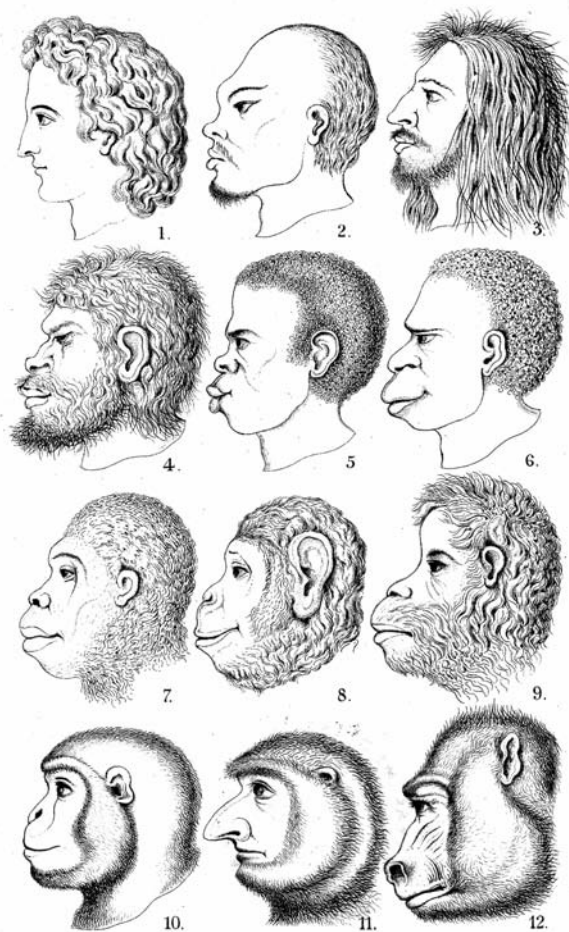
However, the controversy generated by Darwin’s theory was not restricted to its theological implications. It was almost immediately beset by a number of internal problems, and some of the most critical reviews of his work came from fellow scientists. The effects of natural selection operating on the minor variations in characteristics that Darwin postulated required a vast amount of time for the evolution of complex organisms, of the order of 100 million years. Yet calculations based upon the theory of thermodynamics made by the distinguished Victorian physicist William Thompson, Lord Kelvin (1824–1907) in the 1860s, put the age of the earth at only about 30 million years, later revised downward to about 20 million years. This was far too short a period for the operation of Darwinian evolution by natural selection (although later calculations based upon the theory of radioactivity made at the beginning of the 20th century yielded much higher estimates).

Fleeming Jenkin (1833–1885), a Scottish engineer, argued that species could not evolve through the natural selection of minor variations in characteristics given Darwin’s “blending” theory of reproduction (Jenkin, 1867). Favorable variations would be eliminated through dilution within a few generations, as a consequence of organisms possessed of such variations interbreeding with organisms that were not. Jenkin’s objection was not fatal, since as Wallace pointed out, Darwin could have accommodated it by postulating a greater range of variation than he originally allowed (Bowler, 1989). However, Darwin’s blending theory of reproduction was dismissed as implausible by many biologists.

It was also inconsistent with Weismann's theory of the germ-plasm. Darwin did not endorse Weismann's theory, but its very appearance inclined many biologists to oppose the theory of evolution by natural selection. This was because they recognized that Weismann's hard theory of heredity undermined the still-favored conception of evolution as a progressive developmental process based upon the inheritance of acquired characteristics and entailed that natural selection is the *only* viable mechanism for evolution.

Many biologists continued to endorse progressive developmental theories, including neo-Lamarckian theories based upon the inheritance of acquired characteristics. Ernst Haeckel (1834–1919), the foremost German “advocate” of Darwinian theory, treated the individual development of the embryo as the model for the progressive development of species (as had earlier developmental theorists such as Robert Chalmers). Haeckel's **recapitulation theory**, according to which **ontogeny** (the growth of individual organisms) recapitulates **phylogeny** (the evolutionary history of a species), was based upon a teleological conception of evolution that represented humankind—or at least those white European races identified as the highest forms of humankind—as the pinnacle of evolutionary progress (Haeckel, 1876).

Darwin resisted such a conception (although he endorsed the theory of recapitulation), given his vision of evolution as an irregularly branching tree. Yet he failed to convince most of his fellow scientists as well as his theological critics. The late 19th century saw the “eclipse of Darwinism” (Bowler, 1983), and by the end of the century Darwin's theory was dismissed as a historical curiosity (Boakes, 1984). Most theorists continued to hold progressive developmental theories of evolution based upon the inheritance of acquired characteristics. It was only when experimental biologists turned their attention to the postulated biological mechanisms for the inheritance of acquired characteristics at the turn of the century and found them wanting that the climate began to shift again in favor of Darwin's theory of natural selection (Bowler, 1989).



Haeckel's representation of the white European race as the pinnacle of human evolutionary progress.

The Descent of Man

Darwin had studiously avoided discussion of the evolution of humans in *The Origin of Species*, although others (such as Bishop Wilberforce) had been quick to make the connection between apes and men. *The Origin of Species* contained only a single sentence addressed to the question, which promised that the theory of natural selection would throw light upon the evolution of the human species. Darwin was eventually forced to address the issue in *The Descent of Man* (1871), after Huxley drew attention to similarities between the brains of apes and humans in *Evidence as to Man's Place in Nature* (1863), and Wallace published two papers in which he denied that natural selection could account for the emergence of distinctive human characteristics.

In "The Origin of Human Races and the Antiquity of Man Deduced From 'The Theory Of Natural Selection'" (1864), Wallace claimed that although natural selection could account for the origin of the human species, the human brain had developed to such a degree that humans were able to surmount the mechanism of natural selection. According to Wallace, their superior intelligence, manifested by their ability to create fire, clothing, tools, and shelter and to form cooperative social arrangements, enabled humans to survive changes in climate and habitat. He claimed that human evolution had come to be based upon the cultural accumulation of knowledge rather than natural selection (Boakes, 1984). In "Geographical Climates and the Origin of Species" (1869), Wallace argued that many of the distinctive characteristics of humans, such as their capacity for language, logic, mathematics, music, and art, confer no advantage in the struggle for existence. He claimed that these characteristics could not have been developed through evolution by natural selection. They are so advanced of the needs of humans that they could have been produced only by the intervention of a higher (divine) intelligence (Boakes, 1984).

This proved too much for Darwin, who once again was moved by Wallace to publish his own theory. In *The Descent of Man and Selection in Relation to Sex* (1871), he argued that humans, like animals, are descended from simpler ancestral organisms through natural selection. He consequently maintained that there is strong continuity between human and animal psychology and behavior. According to Darwin, human forms of psychology and behavior are elaborations of the forms of psychology and behavior to be found in animals:

The difference in mind between man and the higher animals, great as it is, is certainly one of degree and not of kind. We have seen that the senses and intuitions, the various emotions and faculties, such as love, memory, attention, curiosity, imitation, reason, &c., of which man boasts, may be found in an incipient, or even sometimes in a well-developed condition, in the lower animals.

—(1871, p. 105)

Darwin held that differences between human and animal psychology and behavior are a product of their different degrees of intelligence. The distinctive characteristics of humans, such as their capacity for language, are a by-product of their superior intelligence, but are strongly continuous with evolutionary precursors in the animal kingdom, such as birdsong and the calls of chimpanzees. Darwin did not believe that humans are descended from apes (although he was popularly represented as having done so by his critics) but did claim that primates and humans are descended from a common ancestor.

Yet in order to explain the vastly superior levels of intelligence to be found in humans as opposed to even the highest primates, and the apparent irrelevance of many distinctive human characteristics to survival, Darwin was forced to downplay the role that natural selection played in human evolution. He suggested that there could be rapid leaps in evolutionary development generated by the inheritance of characteristics acquired in special environments that encourage their exercise, such as human social groupings. He also appealed to sexual selection in the explanation of distinctive human capacities that appear to have no survival value, likening the artistic productions of humans to the magnificent but apparently useless plumage of the male peacock (Boakes, 1984).

The *Times* of London predictably responded to the publication of *The Descent of Man* with an editorial claiming that the acceptance of Darwin's views about the origin of humanity would lead to the collapse of morality, echoing Descartes' fears about the consequences of explaining human behavior in mechanistic terms. Yet despite the controversy generated by his theories, Darwin was generally venerated as a distinguished scientist and model of Victorian propriety. When he died in 1882, he was honored by the nation with a burial in Westminster Abbey (where he is interred close to Isaac Newton).

Darwin's commitment to strong continuity between human and animal psychology may have been partly based upon his belief that nature does not move by leaps ("natura non facit saltum"):

As natural selection acts solely by accumulating slight, successive, favorable variations, it can produce no great or sudden modifications; it can act only by short and slow steps.

—(1859, p. 471)

Since evolution by natural selection operates on minor modifications to heritable characteristics over long periods of time, the evolutionary development of a species can be represented as a continuous series of variations of the original characteristics of ancestral organisms. For Darwin this meant that evolutionary antecedents of human characteristics such as language could be identified in related animal species and early human development, such as birdsong and the babbling of infants.

Yet continuous gradation within the process of evolution by natural selection does not entail strong continuity between human and animal psychology and behavior, any more than the continuous gradation between the states of molecules transformed from a liquid to a gas entails the identity of the properties of liquids and gases. In any case, Darwin abandoned continuous gradation within human evolution when he appealed to rapid leaps in the development of human characteristics to explain the large differences in intelligence between humans and even the highest primates.

Darwin recognized that strong continuity between human and animal psychology and behavior did not follow from the theory of evolution by natural selection. He acknowledged that it was possible that “certain powers, such as self-consciousness, abstraction, &c, are peculiar to man” (1871, p. 105) and tried to provide independent evidence for precursors of human language and rationality in the animal kingdom. Yet the evidence Darwin provided was rather weak. Much of it was anecdotal and intuitive and far less rigorous than the detailed fieldwork and experimental studies reported in *The Origin of Species* and *The Variation of Animals and Plants Under Domestication* (1868/1896). His characterization of the calls of chimpanzees as primitive forms of language was based upon reports by foreign correspondents, and he simply assumed that most people would agree that “animals possess some power of reasoning,” since they “may constantly be seen to pause, deliberate and resolve” (1871, p. 46).

In affirming strong continuity between human and animal psychology and behavior, Darwin affirmed more than was required for a defense of the theory of evolution by natural selection in relation to man. All that was required for the demonstration of a common ancestry with animals was the re-identification in animals of *some* aspects of human psychology and behavior, such as associative learning and instinctual behavior. It was not necessary to demonstrate attenuated forms of human language and rationality in animals. As one of Darwin’s eulogists remarked:

[Mr Darwin] was so anxious to show that the moral life of man is but an evolution from the moral life of the lower animals, that he tried to explain that evolution in a false sense, as if the higher phrase involves nothing that is not found in the lower phrase.

—(Hutton, 1882/1894, p. 145)

Darwinism, Racism, and Sexism

Many 19th-century theorists saw Darwin’s account of evolution by natural selection as a justification of their racist views about the intellectual and social underdevelopment of non-Caucasian races. Haeckel represented Caucasians as the apex

of human development, just as he represented the human race as the apex of evolutionary progress:

The immense superiority that the white race has won over the other races in the struggle for existence is due to Natural Selection, the key to all advance in culture, to all so-called history, as it is the key to the origin of species in the kingdom of the living. That superiority will, without doubt become more and more marked in the future, so that still fewer races of man will be able, as time advances, to contend with the white in the struggle for existence.

—(1883, p. 85)

Imperialists appealed to Darwin's theory in their justification of foreign wars of conquest and colonial expansion.

Yet such views were no more consequences of Darwin's theory of evolution than Spencer's laissez-faire form of social Darwinism. Darwin repudiated the idea that the level of intellectual and social development of other races was fixed by natural selection. Although he was horrified by the appearance and behavior of the natives of Tierra del Fuego that were taken aboard the *Beagle*, he was also impressed by their ability to learn Spanish and English and to adopt European habits and manners (Boakes, 1984). Despite his own advocacy of laissez-faire social Darwinism, Spencer repudiated the imperialism of Victorian Britain (Hawkins, 1997).

Late-19th- and early-20th-century theorists generally appealed to Darwin's theory in support of their *independently* held views on race and imperialism. As Stephen Jay Gould (1980) has noted, one implication of Haeckel's recapitulation theory was that "lower races" would manifest the infantile traits of the superior white race: Adult negroes, for example, would manifest the traits of white children. D. G. Brighton claimed that

The adult who retains the more numerous fetal, infantile traits . . . is unquestionably inferior to him whose development has progressed beyond them. Measured by these criteria, the European or white race stands at the head of the list, the African or negro at its foot.

—(1890, cited in Gould, 1980, p. 214)

Racist supporters of Haeckel's theory found multiple evidences of juvenile traits in the art and superstitious behavior of negroes and other "primitive" races. When the development of genetics undermined the theory of recapitulation and evolutionary theorists argued that humans evolved by retaining the juvenile traits of their ancestors, a process known as **neoteny**, racists simply reversed their position. They maintained the superiority of the white race by appeal to the greater retention of juvenile traits by the white race and quickly discovered multiple evidences of juvenile traits in the "child-like vivacity" of Europeans.

Gould (1980) also noted that supporters of neoteny tended to play down one implication of the theory: Since women are more childlike in their anatomy than men, they are superior to men in terms of evolutionary development. Yet late-19th-century and early-20th-century theorists continued to maintain the constitutional inferiority of women. E. H. Clark, professor of physiology at Harvard, claimed that menstruation exerts such a strain on female physiology that the demands of academic study would pose a danger to female health (Birke, 1986). Granville Stanley Hall (1844–1944), who founded the first laboratory and PhD program in psychology in America at Johns Hopkins University, was opposed to the admission of women to tertiary education, because he maintained that it would interfere with their biologically determined childbearing function (Shields, 1975). Hugo Münsterberg (1863–1916), who took over from William James as director of the Harvard laboratory and psychology program, argued that women should be disbarred from juries and denied the vote because of their constitutional irrationality (Hale, 1980).

Early women psychologists challenged these positions. Helen Bradford Thompson (1874–1947), later Helen Woolley, conducted the first systematic empirical study of sensory-motor and perceptual-cognitive differences between men and women at the University of Chicago and concluded that most differences between men and women are the product of social development (Thompson, 1903). She complained that most of the previous research on sex differences was based upon “personal bias, prejudice and sentimental rot and drivel” (Woolley, 1910). Leta Stetter Hollingworth (1886–1939), who received her doctorate in psychology from Columbia University in 1916, also challenged hereditarian assumptions about the intellectual inferiority of women. In her dissertation research at Columbia, she found that menstruation does not diminish the mental capacities of women (Hollingworth, 1914). Like Thompson-Woolley, she complained that common beliefs about the intellectual inferiority of women were social myths unsupported by evidence (Hollingworth, 1940).

Neo-Darwinism

The fortunes of Darwin’s theory revived in the 20th century. However, the version that is generally accepted today is not Darwin’s original theory, but **neo-Darwinian theory**. George J. Romanes (1848–1894) coined the term to characterize any theory of evolution that represents the mechanism of natural selection as *sufficient* to account for the evolution of species. This was the position adopted by Wallace (1858) and Weissman (1893b), but not by Darwin himself. Although Darwin did maintain that natural selection is in principle sufficient to account for the evolution of species, he also claimed that the inheritance of acquired characteristics plays a role (especially in human evolution). Wallace and Weissman, who were committed to a hard theory of heredity, denied the inheritance of acquired characteristics.

Twentieth-century neo-Darwinian theory is the synthesis of Darwin's theory of natural selection and the modern theory of genetics, which is a hard theory of heredity that precludes the inheritance of acquired characteristics. According to this theory, the mechanisms of variation and inheritance are independent of any adaptive responses made by organisms to their environment during their lifetimes.

The modern theory of genetics is based upon the work of Gregor Mendel (1822–1884), who identified dominant and recessive “factors” in 1866. Mendel sent a paper documenting his results to Darwin, who apparently never read it (Hearnshaw, 1987). He spent most of his life at a monastery in Brno in Moravia (now part of the Czech Republic), where he conducted his now famous researches on honeybees and peas. His results were published in the local *Proceedings of the Natural History Society* in 1866, but remained largely unread until his work was rediscovered in the late 1890s. However, the synthesis of genetic theory and the theory of evolution by natural selection did not occur until the 1930s, after experimental biologists eliminated support for soft theories of heredity that allowed for the inheritance of acquired characteristics.

Darwin's Influence on Psychology

Given the doubtful status of Darwin's theory at the end of the 19th century, one has to be careful in assessing Darwin's impact on the development of scientific psychology. Although most psychologists accepted the fact of evolution, not all of them accepted the theory of natural selection. Although early American psychologists were undoubtedly inspired by Darwin's naturalistic treatment of human and animal psychology and behavior, including his contributions to comparative and developmental psychology (Darwin, 1872/1998, 1877), few of them developed forms of evolutionary psychology based upon his theory of natural selection.

Early American functional psychologists often claimed that their psychology was grounded in Darwinian theory. For example, James Rowland Angell, the leader of the movement, claimed that functional psychologists were disposed to the view that human and animal psychology and behavior was “susceptible of explanation in an evolutionary manner” (1909, p. 152). However, functional psychologists focused on the intelligent adaptation of organisms to their environments rather than the evolutionary significance of psychological or behavioral characteristics developed through natural selection (Sohn, 1976). Early American psychologists did not generally embrace systematic explanations of psychological traits and behavior in terms of their survival value (a notable exception was Warren [1918, 1925]). American psychologists only developed these forms of explanation in the late 20th century (Richards, 1987), with the advent of sociobiology (Dawkins, 1978; Wilson, 1975) and evolutionary psychology (Buss, 1999).

Functional and behaviorist psychologists did advance **selectionist theories** of individual learning, according to which behavior is selected by its consequences

for the organism, and often suggested that these were modeled upon Darwin's theory of evolution by natural selection. Yet such theories of individual learning were conceptually independent of Darwin's theory and predated the development of 19th-century theories of evolution. Gay, Whytt, and Hartley advanced selectionist theories of animal and human learning in the 18th century, and Hartley's theory of individual learning provided the model for Erasmus Darwin's theory of the evolution of species in *Zoonomia* (1794–1796). Müller and Bain developed their accounts of “trial-and-error” learning in advance of the publication of Darwin's theory of evolution by natural selection.

However, there are two important respects in which the development of scientific psychology in America may be said to be distinctively Darwinian in orientation. First, many functional and behaviorist psychologists accepted Darwin's commitment to strong continuity between human and animal psychology and behavior and appealed to strong continuity as justification for their generalization of experimentally identified principles of animal learning and behavior to human learning and behavior.

Second, many American psychologists embraced a distinctive implication of Darwin's theory of the evolution of species by natural selection: that evolution does not ensure human perfection or progress. Darwin claimed that evolution is a process driven by blind mechanistic variation and natural selection: It holds no guarantee of perfection or progress. Unlike Spencer, Darwin did not maintain that human evolution left to develop by itself would naturally ensure the best outcome for the human race, especially when the best outcome was defined in terms of the happiness, fulfillment, and moral worth of individuals and social communities: “We must remember that progress is no invariable rule” (1871, p. 177). According to Darwin, evolution by natural selection does not even ensure the survival of the human race, never mind its highest development.

Accordingly, American psychologists were not generally inclined to adopt the laissez-faire approach to the problems of human psychology and society that Spencer advocated. On the contrary, in the spirit of Francis Bacon, they earnestly believed that the fruits of their new scientific discipline could and should be applied to the alleviation of human suffering and the general improvement of the human condition. They almost immediately set about applying the theoretical principles of their new science in the fields of education, industry, and psychological therapy.

FRANCIS GALTON: INDIVIDUAL DIFFERENCES AND EUGENICS

Francis Galton (1822–1911), Darwin's half-cousin, was born near Birmingham, England, and came from a prosperous family (his father was a successful banker). Although a precocious child who could read by the age of 3 and write by the age

of 4, Galton's scholastic record (like Darwin's) was extremely poor. At the age of 16 he was sent to study medicine at Birmingham General Hospital and continued his medical studies at King's College, London. He later transferred to Cambridge University, where he studied mathematics (on Darwin's advice) and received his degree in 1843. However, he abandoned his prospects for a medical career when he came into a large inheritance on the death of his father, which allowed him to pursue his own interests without the encumbrance of salaried employment or business commitments.

Galton had many interests, insatiable curiosity, and a passion for measurement. He pursued an early career as an explorer. From 1845 to 1846 he traveled throughout Egypt, Sudan, and Syria, hoping to discover the source of the Nile and shoot a hippopotamus upon its banks (Boakes, 1984). He explored a large portion of South-West Africa (now Namibia) in 1850, and the published accounts of his travels gained him the Gold Medal of the Royal Geographic Society in 1853. He was elected president of the Royal Geographic Society in 1856 and a fellow of the Royal Society in 1860.

Galton pioneered the scientific study of weather (he introduced the terms "anticyclone," "high," "low," and "front") and the employment of fingerprints in criminal investigation. He invented the teletype printer and did experiments on blood transfusion with different colored rabbits to test Darwin's "blending" theory of reproduction (they provided no support for it). He created a "beauty map" of Britain based upon the number of women of superior appearance he encountered in different towns.

Galton studied boredom, which he measured by degrees of fidgeting; paranoia, which he self-induced and self-observed over a number of days; and the power of prayer, which he judged not to be efficacious. He studied association, imagery, and memory and pioneered the use of questionnaires and word-association tests in *Inquiries Into Human Faculty and Its Development* (1883). He developed measures of sensory acuity, such as the Galton whistle, an instrument that produces high-pitched whistles of different frequency. He was knighted in 1909 and died in 1919.

Individual Differences

Galton's main contribution to psychology was his pioneering study of individual differences, inspired by Darwin's treatment of the chance variation of inherited characteristics in *Origin of Species*. In 1884, Galton set up an **anthropometric laboratory** at the International Health Exhibition in London and collected 12 months of data on about 10,000 people. In 1888, he created a similar laboratory in the Science Galleries of South Kensington Museum, which provided data on around 7,000 people. For a small fee, subjects were tested on a variety of physical



Galton's anthropometric laboratory in Kensington.

and sensory acuity measures: head size, physical strength, visual and auditory acuity, and reaction time. Since Galton claimed that sensory acuity is significantly correlated with intelligence (and that he had demonstrated the correlation), he claimed that sensory acuity is a measure of intelligence. Because he maintained that sensory acuity is largely inherited, he also maintained that intelligence is largely inherited.

Galton was the first person to systematically apply statistics to the study of psychological characteristics. In *Sur L'Homme* (1835/1969), Adolphe Quetelet (1796–1874) had demonstrated that the Gaussian normal probability curve describes the distribution of many biological and social factors, such as body weight, height, and examination grades. Galton maintained that many psychological characteristics, including human intelligence, are similarly distributed. He introduced the median and percentile as measures of central tendency and invented the correlation coefficient to explore the relation between test (and retest) scores accumulated at his anthropometric laboratories (Galton, 1888).

In *Natural Inheritance* (1889), Galton identified the phenomenon of regression toward the mean: the tendency of extreme values of inherited characteristics (such as the size of peas or human intelligence) to move toward the mean value in future generations. Karl Pearson (1857–1936), Galton's protégé at the University of London, developed this work, and in 1896 devised the measure now known

as the Pearson product-moment coefficient of correlation. Pearson called it r in honor of Galton's discovery of regression toward the mean (r being the first letter in *regression*).

Nature and Nurture

Galton rejected the inheritance of acquired characteristics and Darwin's blending theory of reproduction. He embraced a theory of hard heredity analogous to the germ theory developed by Weissman, although he did little empirical research on biological reproduction. Instead, Galton employed his newly developed statistical tools to demonstrate that human characteristics such as intelligence are largely determined by heredity.

In *Hereditary Genius* (1869), *English Men of Science* (1874), and *Natural Inheritance* (1889), Galton claimed that "man's natural abilities are derived by inheritance, under exactly the same limitations as are the form and physical features" (1869, p. 45). He was the first person to discuss the relative contribution of heredity and environment to the determination of human characteristics and popularized the distinction between nature and nurture in *English Men of Science* (1874). He also pioneered the use of twin studies (comparing identical with fraternal twins, raised together and apart) as a means of estimating the respective contributions of heredity and environment (Galton, 1883).

Although his statistical analyses of kinship relations were impressive, the empirical data on which they were based were rather doubtful. Galton's studies of family relations between eminent professional men such as judges and scientists ignored social factors such as wealth and privilege, and his studies of stature, eye color, and artistic propensity were based upon "family records" submitted anonymously by correspondents as entries in a prize-winning competition (Boakes, 1984).

Given his commitment to the hereditarian determination of most human characteristics, Galton was dismissive of utilitarian environmentalists (such as Mill and Sechenov) who optimistically supposed that all human beings are capable of achieving the same levels of intellectual and moral development, given similar education, training, and experience:

I have no patience with the hypothesis . . . that babies are born pretty much alike, and that the sole agencies in creating differences between boy and boy, and man and man, are steady application and moral effort. It is in the most unqualified manner that I object to pretensions of natural equality.

—(1869, p. 12)

Although praised by Darwin, Galton's work was dismissed by many of his contemporaries, who remained committed to the inheritance of acquired characteristics

and the possibilities of education and training. However, Galton initiated the modern debate between hereditarians and environmentalists, and the tide began to turn in favor of hereditarians at the turn of the century.

Eugenics

Like Darwin, Galton recognized that natural selection operating on chance variations in human characteristics does not ensure the evolution of socially desirable characteristics. In “Gregariousness in Cattle and Men” (1872), Galton claimed that gregariousness had been naturally selected because of its survival value in the distant past, but was now an impediment to social progress:

The hereditary taint due to the primaeval barbarism of our race, and maintained by later influences, will have to be bred out of it before our descendants can rise to the position of members of a free and intelligent society.

—(1872, p. 237)

He suggested that society and government should adopt the practice of “artificial selection” that had served as the original model for Darwin’s theory of evolution by natural selection. Just as farmers employ selective breeding to promote the development of desirable characteristics (for farmers) in domestic animals, civilized societies ought to promote the development of socially desirable characteristics in humans through selective breeding.

Galton coined the term **eugenics** (from the Greek for “well-born”) for this form of artificial selection designed to improve “the productivity of the best stock” (1901, p. 663). He claimed that his intelligence tests (based upon measures of sensory acuity) could be employed to select the most intelligent for the purposes of breeding:

Consequently, as it is easy, notwithstanding these limitations, to obtain by careful selection a permanent breed of dogs or horses gifted with peculiar powers of running, or of doing anything else, so it would be quite practicable to produce a highly-gifted race of men by judicious marriages during several consecutive generations.

—(1869, p. 45)

Galton proposed that “highly-gifted” people should be encouraged to breed early and regularly and should be financially supported by the government. He founded the Eugenics Education Society in 1907 and *Eugenics Review* in 1909. He established research fellowships and later a chair in eugenics at University College, London (with money bequeathed in his will). The first holder was Galton’s protégé Karl Pearson, formerly a professor of mathematics. Galton and Pearson established University College, London, as an institutional base for the



Eugenics wedding of the future?

study of statistics, heredity, and eugenics (Boakes, 1984) and launched the journal *Biometrika* (later *Biometrics*) in 1901. After Galton's death in 1911, Pearson and Charles Spearman (1863–1945) continued to promote the cause of eugenics, as did Cyril Burt (1883–1971) and Hans Eysenck (1916–1997) later in the century.

Galton's original concern was with positive eugenics, which encouraged the breeding of the "well-born" through financial incentives. However, eventually Galton and Pearson turned their attention to negative eugenics, which discouraged the breeding of the "ill-born" through institutionalization and sterilization, in response to public concerns about the overbreeding of "idiots and imbeciles."

Darwin had expressed similar concerns in *The Descent of Man*:

With savages, the weak in body and mind are soon eliminated; and those that survive commonly exhibit a vigorous state of health. We civilized men, on the other hand, do

our utmost to check the process of elimination; we build asylums for the imbecile, the maimed and the sick; we institute poor-laws; and our medical men exert their utmost skill to save the life of every one to the last moment. There is reason to believe that vaccination has preserved thousands, who from a weak constitution would formerly have succumbed to small-pox. Thus the weak members of civilized society propagate their kind. No one who has attended to the breeding of domestic animals will doubt that this must be highly injurious to the race of man. It is surprising how soon a want of care, or care wrongly directed, leads to the degeneration of the domestic race; but excepting in the case of man himself, hardly anyone is so ignorant as to allow his worst animals to breed.

—(1871, p. 168)

The English biologist Edwin R. Lankester (1847–1929) avowed that according to Darwin's theory of natural selection, humans "are as likely to degenerate as progress" (1880, p. 60), a prediction that seemed to be confirmed by statistics demonstrating increases in crime, prostitution, alcoholism, tuberculosis, and "feeble-mindedness" among the working class and urban poor (Soloway, 1990). Calls for negative eugenics programs were made by the physiologist John Berry Haycraft (1857–1922) in *Darwinism and Race Progress* (1895) and by the Oxford idealist philosopher Francis Herbert Bradley (1846–1924), who advocated the "social amputation" of the unfit (Bradley, 1894).

Pearson drew attention to the poor levels of health and education among recruits in the British Army during the Boer War (1899–1902), in which a nation of farmers held the mighty British Empire to a stalemate for three years. He claimed that such reduced levels of "national efficiency" posed a threat to national survival (Pearson, 1901). Galton and Pearson recommended the institutionalization and sterilization of the "unfit" to reverse this downward trend. The British Mental Deficiency Act, which required the institutionalization of the mentally retarded, was passed in 1913 (Bowler, 1989).

Similar concerns were expressed in Europe and America. The German Race Hygiene Society was founded in 1905 and the French Eugenics Society in 1912. Charles Davenport (1866–1944), the author of *Eugenics: The Science of Human Improvement by Better Breeding* (1911), founded the Eugenics Records Office at Cold Spring Harbor, New York. Programs of institutionalization and sterilization were introduced in America in the 1920s and with a vengeance in Nazi Germany in the 1930s.

Many of these programs were driven by prevalent elitist, racist, and sexist attitudes rather than scientific evidence. For example, it was generally assumed that alcoholism was an inherited mental disorder rather than a social problem, although there was little evidence for it. The social prejudice of many eugenicists against the weak, the poor, and the unemployed is plain in the following tirade by Major Leonard Darwin (1850–1943), Charles Darwin's son and president of the

British Eugenics Society from 1911 to 1928, who claimed that the aim of negative eugenics is to discourage and decrease the breeding of

the stupid, the careless, the inefficient, the intractable, the idle, the habitual drunkard, as well as those too feeble in body or health to do a good day's work.

—(1928, p. 58, cited in Hawkins, 1997, p. 230)

However, negative eugenics was not mandated by the theory of evolution by natural selection, and Darwin himself rejected such social policies. He claimed that moral compassion is a product of the evolved human capacity for sympathy and that the benefits of “artificial selection” did not justify the intrinsic evil of neglecting or eliminating the weak and the helpless:

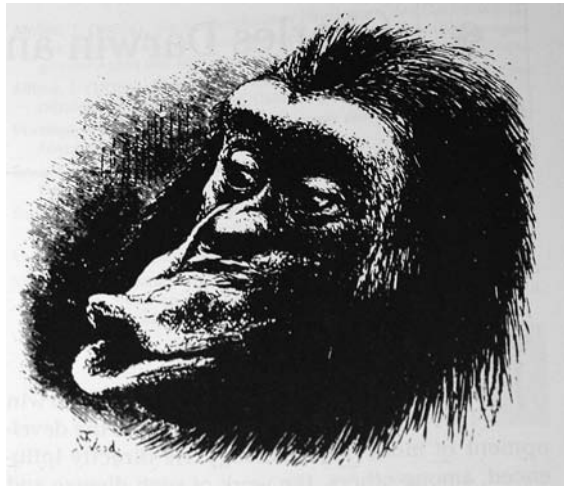
The aid which we feel impelled to give to the helpless is mainly an incidental result of the instinct of sympathy, which was originally acquired as part of the social instincts. . . . Nor could we check our sympathy, even if so urged by hard reason, without deterioration in the noblest part of our nature . . . if we were intentionally to neglect the weak and helpless, it could only be for a contingent benefit, with a certain and great present evil. Hence we must bear without complaining the undoubtedly bad effects of the weak surviving and propagating their kind.

—(1871, pp. 168–169)

MENTAL EVOLUTION AND COMPARATIVE PSYCHOLOGY

The publication of Darwin's *The Descent of Man* (1871) stimulated intense debate about mental evolution in the late 19th century. It also promoted the development of the discipline of comparative psychology, the study of the relation between the psychology and behavior of humans and animals (and humans of different races and cultures). Darwin himself made a significant contribution to the emergent discipline. In *The Expression of Emotions in Man and Animals* (1872/1998), he tried to demonstrate the strong continuity and universality of emotional expression in humans and animals, based upon his own observations (including observations of his infant son) and correspondent reports of the behavior of wild and domestic animals, as well as photographs of actors and the insane.

From the beginning, comparative psychology was linked to developmental psychology via Haeckel's (1876) theory of recapitulation, according to which the development of the individual organism recapitulates the development of the species. As the evolutionary antecedents of human psychology and behavior were held to illuminate their nature, so too were their developmental antecedents. For example, complex human emotions were treated as elaborations of simpler emotions from



Darwin's representation of strong continuity of emotions in humans and animals.

which they were held to have developed, both phylogenetically in humans and animals and ontogenetically in human infants. Darwin also contributed to the fledgling discipline of developmental psychology with the publication of his “Biological Sketch of an Infant” in Bain’s journal *Mind* in 1877. This “child diary” described the early development of Darwin’s own son, William Erasmus Darwin.

A central focus of early comparative and developmental psychology was the question of the degree to which human and animal behavior is determined by instinct. Studies were often designed to investigate whether the early behavioral responses of animals and humans are modifiable by experiential learning.

Spalding on Instinct

Douglas Alexander Spalding (1840–1877) was a (London-born) Scottish roofer whom Alexander Bain took under his wing. Bain arranged for Spalding to attend his lectures at Marischal College, Aberdeen, without having to pay for tuition. Spalding later moved to London, where he qualified as a barrister and worked as a lawyer and teacher. He traveled in Europe in the hope of relieving his tuberculosis (which he had contracted in London), where he met John Stuart Mill, who was living in retirement in Avignon. The two men became close friends, and Mill later arranged for Spalding to serve as tutor to the sons of Lord and Lady Amberley at their country house Ravenscroft in Wales (the youngest of whom was Bertrand Russell, the future philosopher).

The Amberleys were a progressive and liberal couple; Lady Amberley, who was an early champion of female emancipation and birth control, took Spalding into

her bed for his sexual education. Lady Amberley and her daughter died of diphtheria in 1874; when Lord Amberley died two years later, he designated Spalding as legal guardian of his two remaining sons. Lord Amberley's father, Lord Russell, the former prime minister, went to court and broke the will, on the grounds of Spalding's acknowledged atheism and materialism. Spalding was probably the original source of Thomas Huxley's theory of "conscious automata" (Gray, 1968). He died in France of tuberculosis in 1877.

While at Ravenscroft, Spalding conducted a variety of experiments on animal behavior, with Lady Amberley and her children serving as eager research assistants (Boakes, 1984), in which he varied the conditions under which newborn chicks develop (Spalding, 1872, 1873). Working with fertilized chicken eggs, he removed parts of the shell and placed hoods over the eyes of the chicks, to deprive them of visual stimulation:

The conditions under which the little victims of human curiosity were first permitted to see the light were then carefully prepared.

—(1873, p. 283)

Spalding's method of "sensory isolation" also involved inserting wax in the ears of the chicks to deprive them of auditory stimulation. His studies indicated that some responses are instinctual, such as pecking behavior toward insects and directed responses to the calls of the mother hen, whereas other responses are modifiable by experience, such as the chicks' initial tendency to peck at their own excrement (Boakes, 1984).

He also found that certain "imperfect instincts" such as maternal attachment require a critical learning period. According to Spalding, animals "forget" these instincts if they are never practiced: For example, chicks raised for 10 days without exposure to the sounds of other chicks do not subsequently orient themselves to their mother's call. Spalding also identified the form of learning Konrad Lorenz (1935) later characterized as *imprinting*: He noted that newborn chicks would follow the first moving object they experienced, whether it be the mother hen, a dog, or Spalding himself. Mill and Darwin praised Spalding's animal studies, and William James (1890) acknowledged "Mr. Spalding's wonderful article on instinct" in his *Principles of Psychology* (Reed, 1997).

George John Romanes: Animal Intelligence

George John Romanes (1848–1894) abandoned his original enthusiasm for religion after reading Darwin's *Origin of Species* and *Descent of Man* and developed an interest in the theory of mental evolution. The two men formed a close friendship after Romanes visited the aging naturalist at his home in 1874. Darwin informally

designated Romanes as his successor in the emerging field of comparative psychology by bequeathing his notebooks on animal behavior to his young disciple. Romanes studied physiology at Cambridge University; but, as in the case of Darwin and Galton, his private income allowed him to devote his time to his scientific pursuits without having to worry about a profession or business. He built a private laboratory in his native Edinburgh, where he served for a period as a part-time lecturer at Edinburgh University. His experimental studies of reflexive behavior in jellyfish established his scientific reputation, and Romanes was elected to the Royal Society at the age of 31 (Boakes, 1984).

As his own interest in mental evolution developed, Romanes assembled an extensive collection of reports of animal behavior. Many of these were solicited by advertisements in the London *Times*. Others were gleaned from scientific and popular journals or communicated to him through his extensive network of international correspondents. These reports became the basis of his first book on *Animal Intelligence*, published in 1882, which was restricted to the documentation of evidence for mental evolution in animals and men. Romanes later developed his own theories of mental evolution in *Mental Evolution in Animals* (1884a) and *Mental Evolution in Man* (1885).

Like Darwin, Romanes was committed to strong continuity between the psychology and behavior of humans and animals. He claimed that there is “psychological, no less than physiological continuity extending throughout the length and breadth of the animal kingdom” (1882/1977, p. 113). Despite his reverence for Darwin, Romanes ignored his caution against talking of “higher” and “lower” species and represented humankind as the pinnacle of an evolved natural hierarchy. In his own version of the theory of recapitulation, Romanes claimed that the mental capacities of human babies develop through stages equivalent to those of insect larvae around 3 months, to those of reptiles at around 4 months, and to those of dogs at around 15 months (Boakes, 1984).

Romanes’s Methodology Romanes’s own theories of mental evolution were quickly forgotten by later generations of comparative psychologists. His primary influence on the history of psychology was entirely negative, through the vigorous rejection of his methodology and data by animal and behaviorist psychologists. He was roundly condemned for his use of **introspection by analogy**, his attempt to understand animal mentality on the basis of analogies with human mentality accessible to introspection, and for his use of **anecdotal data**, based upon secondhand reports of animal behavior. Romanes was also condemned for his **anthropomorphism**: his ascription of human mental capacities such as abstract reasoning to animals.

Like earlier theorists such as John Stuart Mill (1865), Romanes claimed that we have direct and certain introspective knowledge of our own mental states, but have only indirect and uncertain knowledge of the mental states of other people and animals through inference from their observed behavior, based upon analogies

with our own mental states and behavior. For example, although we can introspect our own pain when we step on a sharp stone and automatically withdraw our foot and scream, we can only infer that other people or animals are in pain when we observe them behaving in similar ways in similar circumstances:

For if we contemplate our own mind, we have an immediate cognizance of a certain flow of thoughts or feelings. . . . But if we contemplate mind in other persons or organisms, we have no such immediate cognizance of their thoughts or feelings. In such cases we can only *infer* the existence and nature of thoughts and feelings from the activities of the organisms which appear to exhibit them. . . . Starting from what I know subjectively of the operations of my own individual mind, and the activities which in my own organism they prompt, I proceed by analogy to infer from the observable activities of other organisms what are the mental operations that underderlie them.

—(1882/1977, pp. 1–2)

There was nothing inherently unscientific about Romanes's ascription of mentality and consciousness to animals on the basis of analogical inference. A great many theoretical inferences in science are based upon postulated analogies with the properties of observable entities. The wave theory of light was based upon a postulated analogy between the properties of light and the properties of ocean waves, and the Bohr theory of the atom was based upon a postulated analogy between the properties of atoms and planetary systems. The scientific adequacy of such theories does not depend upon their conjectural origin, but on the quality of observational and experimental evidence garnered in support of them.

Unfortunately the quality of the evidence garnered by Romanes in support of his attributions of mentality and consciousness to animals was rather poor. Most of it was based upon anecdotal reports of animal behavior, gleaned from responses to his newspaper solicitations or from overseas correspondents—although in this respect his *Animal Intelligence* was no different from Darwin's *Descent of Man*, and Romanes did include John Lubbock's (1834–1913) early experimental studies of ant behavior (Lubbock, 1882). These reports documented isolated instances of animal behavior that were apparently conscious and intelligent, but were doubtfully representative of the behavior of the relevant species. As Edward L. Thorndike (1874–1949), one of the pioneers of American experimental animal psychology, later complained, such anecdotal reports of animal behavior are generally unrepresentative because they describe unusual or surprising behavior:

Dogs get lost hundreds of times and no one ever notices it or sends an account of it to a scientific magazine. But let one find his way from Brooklyn to Yonkers and the fact immediately becomes a circulating anecdote. Thousands of cats on thousands of occasions sit helplessly yowling, and no one takes thought of it or writes to his friend, the professor; but let one cat claw at the knob of a door supposedly as a signal to be

let out, and straightway this cat becomes the representative of the cat-mind in all the books. . . . In short, the anecdotes give really the *abnormal* or *supernormal* psychology of animals.

—(1911, pp. 24–25)

Thorndike and later behaviorist psychologists maintained that the scientific study of animal behavior should be based upon controlled experimental studies of animal behavior.

Romanes recognized the methodological weaknesses of anecdotal reports, which is why he tried to assemble such a wide range of naturalistic reports of animal behavior in *Animal Behavior*. Ironically, his main reason for assembling these reports was to improve upon the “works of anecdote mongers,” by including only observations reported by “trustworthy” authorities. Unfortunately, his criteria for trustworthy authority tended to be more social than scientific. He included many reports of animal behavior because they were supported by “competent” judges such as bishops, major-generals, and well-bred young ladies.

Although he was an accomplished experimental physiologist, Romanes did not develop a program of experimental research on animal behavior and in fact conducted few empirical studies. He followed Darwin’s suggestion that he procure a monkey for study, but entrusted its care to his sister (Boakes, 1984). He seems to have supposed that experimental studies are unnecessary in comparative psychology, since the direct introspection of mental states provides the basis of inferential knowledge in both human and animal psychology:

In the science of psychology nearly all of the considerable advances which have been made, have been made, not by experiment, but by observing mental phenomena and reasoning from these phenomena.

—(1884a, p. 12)

This claim did not recommend Romanes to later behaviorist psychologists, who rejected the introspective analysis of mental states and focused on the experimental study of animal and human behavior. However, their rejection of introspection and anecdotal evidence did not justify their almost exclusive commitment to experimentation. Although Romanes was wrong to base his theories of mental evolution on anecdotal reports, he was not wrong to base his theories on naturalistic as opposed to experimental studies of animal behavior. Naturalistic observations in open systems (not subject to experimental control) are commonplace in many sciences, and the only type of observational evidence available in many venerated physical sciences such as geology and astronomy. They have also been the staple diet of students of animal behavior from Aristotle to Niko Tinbergen (1907–1988) and Konrad Lorenz (1903–1989), who received the Nobel Prize in biology in 1973 for their **ethological** studies of the behavior of animals in their natural environment.

Although there was nothing inherently unscientific about Romanes's attribution of human forms of mentality and consciousness to animals, many of his attributions of mentality and consciousness to animals were excessively anthropomorphic. He ascribed complex human emotions such as hypocrisy to dogs and abstract knowledge of mathematical principles to monkeys. He solemnly reported the following instance of the trial of a crow, communicated to him by Major-General Sir George Le Grand Jacob, who made the observation during his service in India:

Soon a gathering of crows from all quarters took place, until the roof of the guard house was blackened by them. Thereupon a prodigious clatter ensued; it was plain that a "palaver" [discussion] was going forward. . . . After much cawing and clamour, the whole group suddenly rose into the air, and kept circling round a half-dozen of their fellows, one of whom had been clearly told off for punishment, for the five repeatedly attacked it in quick succession, allowing no opportunity for their victim to escape, which he was clearly trying to do, until they cast him fluttering on the ground about thirty yards from my chair.

—(1882/1977, pp. 324–325)

Although there is nothing inherently unscientific about a comparative psychology in which forms of human psychology and behavior are attributed to animals by analogical inference, based upon naturalistic observations of animal behavior, Romanes's excessive anthropomorphism and reliance on anecdotal data gave this form of comparative psychology a negative reputation that it took almost 100 years to recover from. With the rejection of Romanes's theoretical analogies and naturalistic observations, the scientific psychological study of animal mentality and consciousness did not revive until the 1970s (Griffin, 1976), in the wake of the general "cognitive revolution" in psychology.

Conwy Lloyd Morgan: Morgan's Canon and Emergent Evolution

Darwin had informally designated Romanes as his successor in the developing field of comparative psychology. Romanes did the same for Conwy Lloyd Morgan (1852–1936), whom he appointed as his literary executor. He considered Morgan to be an astute observer of animal behavior. Romanes had expressed some skepticism in *Animal Intelligence* about reports of scorpions being prone to suicide and was impressed by Morgan's report (1883) of some (rather cruel) experiments in which he had demonstrated that scorpions could not be induced to commit suicide even under conditions of extreme stress (Boakes, 1984). Morgan acknowledged Romanes's role in the early development of comparative psychology in his 1890 book *Animal Life and Intelligence*. Yet in contrast to Romanes, who revered Darwin and his work, Morgan adopted an increasingly critical attitude to Romanes's theories and methods.

Morgan originally trained as a mining engineer at the London School of Mines (later the Imperial College of Science and Technology). Thomas Huxley, who was a lecturer there at the time, persuaded him to study animal behavior instead. In 1884, after years of travel and part-time employment, Morgan secured the position of professor of geology and zoology at Bristol College, which later became the University of Bristol. Although he had hoped to establish a research institute and chair in comparative psychology, he was unsuccessful in his efforts, even when he was appointed vice-chancellor of the University of Bristol. He visited the United States in 1896 and lectured in Boston, Chicago, and New York. His first book was *Animal Life and Intelligence* (1890). It was followed by *Introduction to Comparative Psychology* (1894), *Habit and Instinct* (1896), which was based upon his American lectures, and *Animal Behavior* (1900). Morgan was elected a fellow of the Royal Society in 1899.

Morgan's Canon Morgan is famous in the history of psychology for his formulation of the methodological principle that has come to be known as Morgan's canon. According to this principle, psychologists should eschew explanations of animal behavior in terms of complex cognitive states when simpler explanations will suffice, such as in terms of instincts or learned habits. In *Introduction to Comparative Psychology* Morgan claimed that

In no case may we interpret an action as the outcome of the exercise of a higher psychical faculty, if it can be interpreted as the outcome of the exercise of one which stands lower in the psychological scale.

—(1894/1977, p. 53)

Formally considered, Morgan's canon was simply a restatement of the scientific principle of simplicity, sometimes known as "Occam's razor": When two or more competing theories are equivalent in terms of their explanatory and predictive success (when they can accommodate the same range of empirical data), the simplest theory should be preferred over the more complex. In the case of comparative psychology, simpler explanations of animal behavior in terms of reflexes, instincts, or learned habits should be preferred over more complex explanations in terms of the comprehension of abstract mechanical principles, when they accommodate the same range of animal behavior. Wundt expressed a similar principle (the "law of parsimony") in his *Lectures on Human and Animal Psychology*, which only allowed "recourse to be had to complex principles of explanation when the simplest ones have proved inadequate" (1863/1894, p. 350).

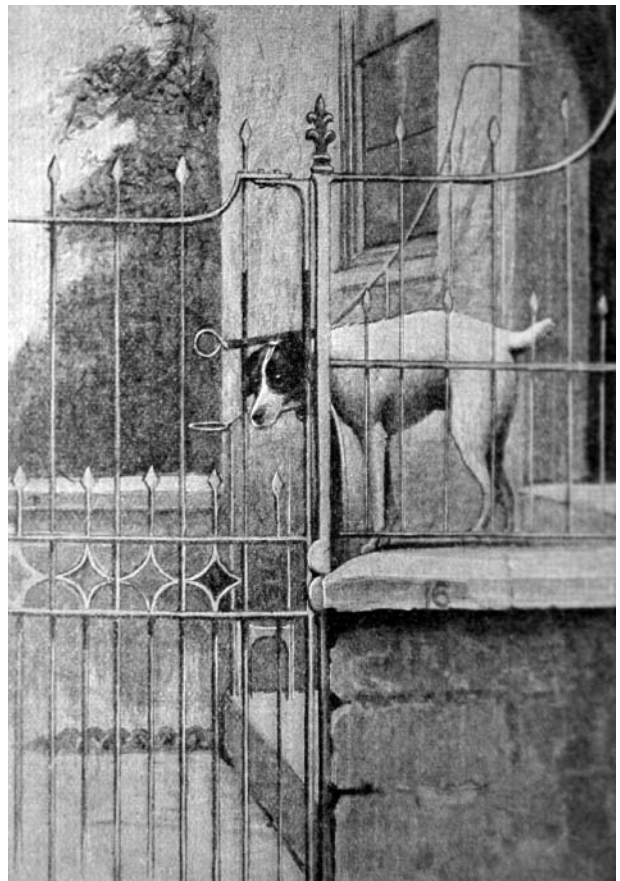
Morgan believed that "a very large percentage of the activities of animals can be fairly explained as due to intelligent adaptation through association founded on sense-experience" (1894/1977, p. 358). For example, Romanes had explained the ability of cats and dogs to open gate and door latches in terms of their rational

comprehension of mechanical principles (1884a, p. 193), but Morgan suggested that such behavior could be adequately explained as forms of “trial and error learning with accidental success”—as instances of the Spencer-Bain principle.

To illustrate his point, Morgan recounted how his fox terrier Toby had learned to open the latch on his courtyard gate:

I watched from the first the development of the habit. The facts are as follows: I may premise that the gate is of iron, and has iron bars running vertically with interspaces of five or six inches between. On either side is a wall or low parapet, on which are similar vertical rails. The latch of the gate is at a level of about a foot above that of the top of a low wall. When it is lifted the gate swings open by its own weight. The gate separates a small garden, of only a few square yards area, from the road. When the dog is put out of the front door he naturally wants to

get out into the road, where there is often much to interest him; cats to be worried; other dogs with whom to establish a sniffing acquaintance, and so forth. . . . He ran up and down the low wall, and put his head out between the iron bars, now here, now there, now elsewhere, anxiously gazing into the road. This he did for quite three or four minutes. At length it so happened that he put his head beneath the latch, which, as I have said, is at a convenient height for his doing so, being about a foot above the level of the wall. The latch was thus lifted. He withdrew his head, and began to look out elsewhere, when he found that the gate was swinging open, and out he bolted. After that, whenever I took him out, I shut the gate in his face, and waited till he opened it for himself and joined me. I did not give him any assistance in any way, but just waited and watched, sometimes taking him back and making him open it again. Gradually he went, after fewer pokings of his head in the wrong place, to the one opening at which the latch was lifted. But it was nearly three weeks



Trial-and-error learning. Lloyd Morgan's fox terrier Toby opening the gate.

from my first noticing his actions from the window before he went at once and with precision to the right place and put his head without any ineffectual fumbling beneath the latch. Even now he always lifts it with the back of his head and not with his muzzle which would be easier for him.

—(1894/1977, pp. 289–290)

Morgan granted that a casual observer of the dog's behavior might reasonably suppose that the dog had "clearly perceived how the end in view was to be gained and the most appropriate means for effecting his purpose" (1894/1977, p. 288). However, he claimed that upon learning how the "clever trick originated," any critical observer would agree that it was unnecessary to attribute rational comprehension of mechanical principles to explain the dog's behavior.

Morgan was especially skeptical of explanations of animal behavior in terms of their perception of "particular relations among phenomena" or as exercises of "conceptual thought," since he believed that animals were incapable of perceiving relations or engaging in conceptual thought. However, Morgan never claimed that it is scientifically illegitimate to offer explanations of animal behavior in terms of complex mentality or consciousness and freely admitted that

there is a small . . . outstanding percentage of cases, the explanation of which seems to involve the attribution to animals of powers of perception and rational thought.

—(1894/1977, p. 358)

He certainly did not believe that explanations of animal behavior in terms of "intelligent adaptation through association founded on sense-experience" should be preferred over explanations in terms of more complex cognitive states just because the former explanations are simpler, since he insisted that "the simplicity of an explanation is no necessary criterion of its truth" (1894/1977, p. 54). As he noted, the theory of the divine creation of species is simpler than the theory of the evolution of species by natural selection but is not preferred by biologists for that reason. Morgan insisted that a more complex cognitive explanation is to be preferred over a simpler explanation in terms of instincts or habits precisely when the evidence warrants such an explanation:

The canon by no means excludes the interpretation of a particular activity in terms of the higher processes, if we already have independent evidence of the occurrence of these higher processes in the animal under observation.

—(1903, p. 59)

Morgan kept an open mind about such matters, merely insisting that questions about levels of cognition and consciousness attributable to animals should be determined by empirical investigation. He was skeptical of many of

the attributions of animal cognition and consciousness advanced by Romanes in *Animal Intelligence* and insisted that hypotheses about animal mentality and consciousness should be based upon the careful observation of animal behavior, ideally under controlled experimental conditions. He claimed that they should be evaluated “not by any number of anecdotes . . . but by carefully conducted experimental observations . . . carried out as far as possible under nicely controlled conditions” (1894/1977, p. 359).

However, Morgan did not reject naturalistic observation and was sensitive to the potential distortion of animal behavior created by experimental manipulation and control. Although he agreed with Thorndike’s explanation of “trial and error” learning in terms of the “law of effect” (Thorndike, 1911), he was highly critical of the artificial experimental studies on which it was based, in which animals learned to escape from specially constructed “puzzle-boxes” (Morgan, 1898).

Morgan was also critical of Romanes’s excessive anthropomorphism. Yet he did not object to the anthropomorphic interpretation of animal behavior per se (Costall, 1993), since he believed this was an integral feature of comparative psychology:

Our psychological interpretations are invariably anthropomorphic. All we can hope to do is to reduce our anthropomorphic conclusions to their simplest expression.

—(1900, p. 48)

Morgan introduced his canon to bring some measure of scientific objectivity to the theoretical attribution of mentality and consciousness to animals, but never intended it as a prohibition against such attributions. However, it did present a challenge to later animal psychologists to develop explanations of animal behavior without reference to mentality or consciousness. Thorndike and Pavlov took up this challenge in the early part of the 20th century (although neither were responding directly to Morgan): They offered explanations of animal behavior that did not appeal to cognition or consciousness (or at least did not appear to) and that were based upon controlled experimental studies of animal behavior. These served as exemplars of theory and method for later generations of behaviorist psychologists.

The conceptual link between Morgan’s cautious approach to the attribution of mentality and consciousness to animals and the restrictive approach of later behaviorist psychologists may be demonstrated by considering Romanes’s response to Morgan’s original doubts about the scientific legitimacy of ascribing mentality and consciousness to animals as opposed to humans. Morgan had questioned whether an objective comparative psychology was possible. He believed that inferences to the mental states of other people on the basis of their behavior were legitimate because they could be confirmed by their verbal reports: For example, other people could affirm the pain or means-end reasoning ascribed to them on the basis of their behavior. However, since this is not possible with respect to

animals, Morgan originally believed that inferences about the mental states of animals were illegitimate and that comparative psychology was restricted to the study of physiology and behavior (Morgan, 1884).

Romanes objected that acceptance of the verbal reports of other people is just another form of inference on the basis of observed behavior: “it is for me nothing more than my own interpretation of a meaning by the observable activities of an organism” (1884a, p. 379). He complained that Morgan could not consistently endorse inferences about the mental states of other humans but not animals, given obvious similarities in their adaptive and intelligent behavior. According to Romanes, inferences about human and animal mentality and consciousness stand or fall together. For, as he presciently noted, any doubts about the legitimacy of explanations of animal behavior in terms of mentality and consciousness based upon inferences from their behavior could be generalized to explanations of human behavior in these terms:

In whatever measure [Morgan] is *on principle* a skeptic touching the inferences which this science [of comparative psychology] is able to draw as to the existence and nature of animal psychology, in that measure I think he ought in consistency also to be a skeptic with reference to the same points in the science of human psychology.

—(1884b, cited in Costall, 1993, p. 119)

This was precisely the skeptical conclusion drawn by the behaviorist John B. Watson in the early decades of the 20th century: There is no need to appeal to cognition or consciousness in the explanation of either animal or human behavior. This conclusion was certainly in accord with the letter of Morgan’s canon, since Morgan agreed that the canon applied to the explanation of both animal and human behavior. Just as Morgan had argued that most or all animal behavior could be explained in terms of “intelligent adaptation through association founded on sense-experience,” so too it could be argued that most or all human behavior could also be explained in these terms.

Later behaviorist psychology was based upon the restrictive employment of Morgan’s canon, initially as a prohibition against explanations of animal behavior in terms of cognition and consciousness and later as a prohibition against the explanation of human behavior in these terms. There is, however, considerable irony in this historical development, since part of the point of Morgan’s canon was to preclude precisely the sorts of generalization of explanations of animal behavior to human behavior that became the hallmark of behaviorist psychology.

Emergent Evolution Attributions of human forms of mentality and consciousness to animals by comparative psychologists were generally based upon the assumption of strong continuity between human and animal psychology and

behavior. Romanes, like Darwin, maintained that evolutionary precursors of the psychology and behavior of humans, such as attenuated forms of the perception of relations, abstract reasoning, and language, could be found in animals. Yet Morgan's canon was based upon the assumption that there may be strong *discontinuity* between human and animal psychology and behavior. According to Morgan, certain forms of human psychology and behavior, such as abstract thought or behavior based upon means-ends reasoning, may not be identifiable *in any form or to any degree* in animals, since no animal may have reached the level of evolutionary development required for the emergence of these psychological and behavioral capacities. Although Morgan constantly stressed that theoretical disputes about animal mentality and consciousness should be determined by empirical and experimental research, and had no problem attributing consciousness to animals, he believed that animals are incapable of even attenuated forms of abstract thought and behavior based upon means-end reasoning. He denied that "any animals have reached that stage of mental evolution at which they are even incipiently rational" (1894/1977, p. 377).

Morgan pointed out that this (tentative) conclusion was entirely consistent with his canon:

It is clear that any animal may be at a state where certain higher faculties have not yet been evolved from their lower precursors; and hence we are logically bound not to assume the existence of these higher faculties until good reasons have been shown for such existence.

—(1894/1977, p. 59)

Morgan's own skeptical doubts about the explanation of animal behavior in terms of human capacities such as abstract thought and means-end reasoning were predicated on the assumption that *humans have these capacities but animals do not*. He claimed that if the evidence did not support the attribution of even attenuated forms of abstract thought or means-end reasoning to animals, then we should abandon the assumption of strong continuity and recognize that there may be a "radical difference" between the psychology and behavior of humans and animals:

Are there apparent breaches of continuity in mental development? I am disposed to answer that such apparent breaches there are. The step from mere sentience to consentience probably involved such a breach or new departure in the development curve. The step from consentience, or sense-experience, to reflection and thought certainly involves, in my judgement such a new departure. . . .

If the dividing line between sense-experience and reflection is to be drawn between the lower animals and man, then we may say that there is a breach of

continuity of development at this stage of evolution analogous to the breach of continuity between the organic and inorganic stages of development.

—(1894/1977, pp. 354–355)

Morgan claimed that some human capacities, such as the perception of relations and abstract thought, are not to be found in animals, even in incipient or attenuated form. He did not deny that such capacities are products of evolution, but maintained that animals had not reached a stage of evolutionary development in which these capacities had emerged.

Morgan's view that distinctive human capacities such as the perception of relations and abstract thought are forms of psychology that emerge at more complex levels of biological organization and development, which came to be known as **emergent evolutionism**, was the evolutionary analogue of Mill's principle of mental chemistry. There was nothing especially radical about Morgan's view, which, as he correctly maintained (and Darwin had earlier acknowledged) was entirely consistent with the theory of evolution by natural selection.

While he acknowledged continuities in reflexive behavior, instinct, and habit learning between humans and animals, Morgan believed it was premature to assume that animals had reached a stage of evolution in which capacities such as the perception of relations and abstract thought had emerged. For Morgan, this meant that although some forms of explanation in terms of associative processes could be applied to both animal and human behavior, some forms of explanation in terms of cognition and consciousness could be applied only to human behavior. Some human behavior but no animal behavior could be explained in terms of the perception of relations and abstract thought.

This is precisely what was denied by later generations of animal and behaviorist psychologists, who remained committed to the principle of strong continuity. Yet while later biologists and animal psychologists embraced the principle of strong continuity advocated by Romanes and Darwin, they *reversed* the explanatory direction of comparative psychology. Instead of following Romanes and Darwin in trying to explain animal psychology and behavior as attenuated approximations of the highest forms of human cognition and consciousness, later biologists and animal and behaviorist psychologists followed Spencer in explaining all human psychology and behavior in terms of the elaboration of basic associative and reflexive processes to be found at even the lowest levels of animal life.

Jacques Loeb (1859–1924), professor of biology at the University of Chicago, who taught Watson biology and physiology, tried to account for all forms of psychology and behavior in terms of basic associative processes. He maintained that the higher cognitive capacities of humans and animals are elaborations of associative memory, which is itself an elaboration of more basic associative mechanisms underlying reflexes and tropisms (the automatic, mechanical orientation

of plants and animals to light and gravity). It was this type of associative theory that was generalized to the explanation of all human behavior by later behaviorist psychologists, on the basis of experimental studies of the behavior of cats, dogs, rats, and pigeons.

STIMULUS-RESPONSE PSYCHOLOGY

This was the legacy for early-20th-century psychology of the debates about mental evolution within the emerging discipline of comparative psychology. Although the theory of evolution by natural selection did not mandate commitment to strong continuity between human and animal psychology and behavior, this was the position adopted by most 20th-century functionalist and behaviorist psychologists.

This commitment was reinforced by the common commitment to the strong continuity between higher cognitive and lower sensory-motor reflexive processes presupposed by the sensory-motor theory of the nervous system. When every cognitive function was held to be sensory-motor in nature, it was natural to presume that human cognitive functions were merely elaborations of the basic sensory-motor functions to be found in animals.

This reinforcement was not accidental or unidirectional. Spencer's commitment to the strong continuity of the sensory-motor theory of the nervous system shaped his distinctive conception of evolutionary psychology, with its increasingly more complex levels of association. And neurophysiologists such as John Hughlings Jackson (who was directly influenced by Spencer) embraced the sensory-motor theory of the nervous system because of their commitment to the theory of evolution:

If the doctrine of evolution be true, all nervous centers must be of sensory-motor constitution. *A priori*, it seems reasonable to suppose that, if the highest centers have the same composition as the lower, being, like the lower made up of cells and fibres, they have also the same constitution.

—(Jackson, 1931, 2, p. 63)

These two commitments provided the conceptual foundation for the forms of stimulus-response psychology based upon principles of associative learning that dominated American psychology in the first half of the 20th century. Morgan's reasonable position that some but not all human behavior could be explained in terms of the principles of animal psychology was uniformly rejected by behaviorist psychologists and only resurrected in American psychology after the cognitive revolution of the 1950s. It was, however, the position adopted by Wilhelm Wundt, who founded the discipline of scientific psychology in Germany in the late 19th century.

DISCUSSION QUESTIONS

1. Lamarck and Spencer replaced the extrinsic teleology of a divinely created natural order with the intrinsic teleology of progressive evolutionary development toward a natural order. Is an intrinsic teleological account any more plausible than an extrinsic teleological account?
2. Spencer believed that his form of laissez-faire social Darwinism followed from his theory of evolution based upon the inheritance of acquired characteristics and natural selection. Others, including Darwin, disputed this. Do the principles of the inheritance of acquired characteristics and natural selection have any implications for social policy?
3. Is there really no vestige of purpose, perfection, or progress in Darwin's theory of evolution by natural selection? Is there really no justification for talking about human beings as higher than earthworms?
4. Wallace claimed that distinctively human characteristics such as capacities for mathematics, music, and art could not be explained as a product of natural selection, since they confer no advantage in the struggle for existence. Can such characteristics be explained in terms of the theory of natural selection?
5. Modern genetics and medical science offer far greater possibilities for positive and negative eugenics than could have been envisioned in Galton's time. Have the moral issues remained the same? What would Darwin have thought?
6. Morgan claimed that certain human capacities such as the perception of relations and abstract reasoning are strongly discontinuous with the capacities of other animals. Do you think his position was consistent with the theory of evolution by natural selection and with what was known about the nervous system at the end of the 19th century? Do you think there are human cognitive capacities that cannot be attributed to animals in even incipient form?

GLOSSARY

anecdotal data Data based upon secondhand reports.

anthropometric laboratory Galton's laboratory for the measurement of human characteristics such as sensory acuity and reaction time.

anthropomorphism Ascription of human mental capacities such as abstract reasoning to animals.

blending theory Darwin's theory of reproduction, according to which offspring inherit half of the particles that pass through the adult parents to their reproductive organs.

emergent evolutionism Theory that human capacities such as the perception of relations and abstract thought are distinctive forms of human psychology that emerge at more complex levels of biological organization and development.

ethology Study of the behavior of animals in their natural environment.

eugenics Galton's term (from Greek for "well-born") for programs of artificial selection that encourage or promote the breeding of the "highly gifted" and discourage or prevent the breeding of "idiots and imbeciles."

fluid materialism Theory that the electrical nature of the nervous system is the basis of life and mind.

hard heredity Theory of heredity according to which the mechanism of inheritance through biological reproduction is independent of the life history of organisms.

inheritance of acquired characteristics Doctrine that useful modifications that are made to existing organs through increased use or that are developed in response to environmental pressures during the lifetime of an organism tend to be inherited by future offspring.

introspection by analogy The attempt to understand animal mentality on the basis of analogies with human mentality accessible to introspection.

laissez-faire In economic theory, the doctrine (from French for "leave to do") that governments should not intervene in the market. In Spencer's theory of evolution, the doctrine that government should not intervene to alleviate the condition of the poor, sick, and unemployed and that societal progress is best assured by leaving biological, psychological, and social evolution to take its natural course.

natural selection The selection of variations in inheritable characteristics that are conducive to the survival of a species through the struggle for existence.

neo-Darwinian theory Theory of evolution that represents the mechanism of natural selection as sufficient to account for the evolution of species.

neoteny Theory that humans evolved by retaining the juvenile traits of their ancestors.

ontogeny The growth of individual organisms.

phylogeny The evolutionary history of a species.

Protestant work ethic Doctrine that everyone has a chance to rise in society as long as they make the effort to adapt to changing circumstances and that only those who make the effort deserve to benefit.

recapitulation theory Ernst Haeckel's theory that ontogeny recapitulates phylogeny.

reform Darwinism Doctrine that governments should introduce programs designed to create social conditions that encourage individuals to improve themselves (such as improved public health and education).

- selectionist theory** Theory of individual learning according to which behavior is selected via its consequences for the organism.
- social Darwinism** The application of theories of evolution based upon the survival of the fittest to theories of social change and political practice.
- soft heredity** Theory of heredity according to which the mechanism of inheritance through biological reproduction can be influenced by the life history of organisms.
- struggle for existence** Phrase coined by Spencer to describe competition between members of a population for limited food resources.

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Psychology in Germany

THE ACADEMIC DISCIPLINE OF SCIENTIFIC PSYCHOLOGY WAS FOUNDED institutionally in Germany at the end of the 19th century. It was a natural outgrowth of the progressive German university system, which was hospitable to the development of new disciplines such as linguistics and psychology. Wilhelm Wundt (1832–1920), who founded scientific psychology in Germany in 1879, the year he set up his experimental laboratory at the University of Leipzig, characterized the new discipline as *physiological psychology*. This was not because he believed that psychological states and processes must be reductively explained in terms of physiological states and processes, but because he believed that scientific psychology should appropriate the experimental methods that had proved so successful in the development of 19th-century German physiology.

Wundt's new experimental program attracted many foreign students, including many Americans, who sought to attain professional qualifications in the new discipline. Having mastered the elements of the new psychology and the structure of the German university system, they returned home to create their own laboratories and PhD programs in psychology.

Wundt's distinctive form of scientific psychology was eventually displaced within Germany as rival programs were created at other German universities. As the 20th century advanced, German psychology, which faced increasing opposition from the philosophical community, developed into precisely the type of applied discipline that Wundt feared it would become. In a sense Wundt's fate was like that of the sorcerer's apprentice. In creating a form of scientific psychology based upon laboratory science, he unleashed powerful forces that he was unable to control—forces that, over a few generations, radically transformed the discipline (Danziger, 1990, p. 34).

PSYCHOLOGY IN GERMANY BEFORE WUNDT

Psychology had been recognized as a distinctive field of inquiry long before the creation of Wundt's laboratory, and academic philosophers in various countries had offered courses in the subject. For example, psychology was offered as a course